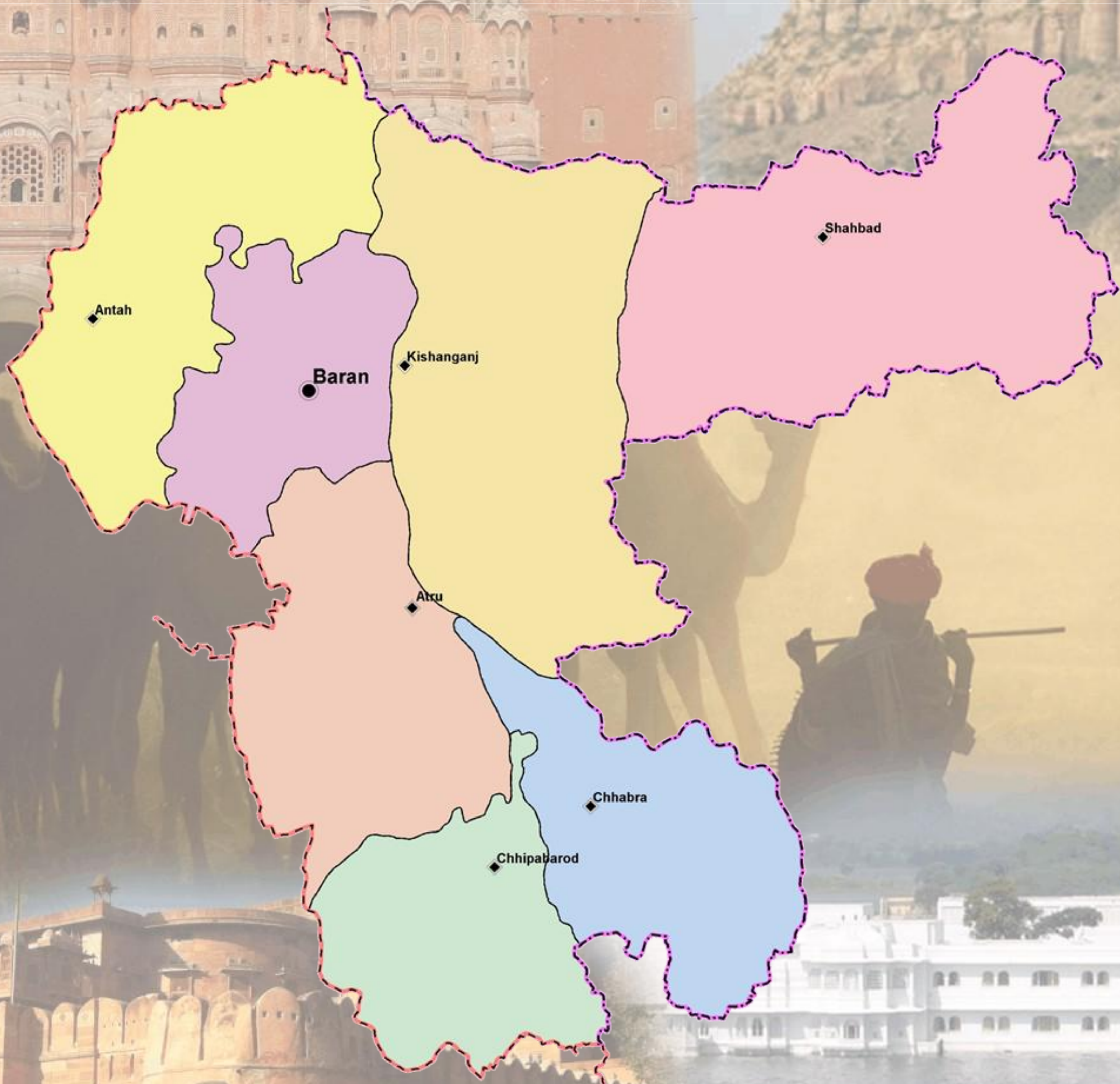
Ground Water Department,
Rajasthan

Hydrogeological Atlas of Rajasthan

Baran District

European Union
State Partnership Programme



2013

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Message

सत्यमेव जयते

Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of 'Hydrogeological Atlas of Rajasthan'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.

(Ashok Gehlot)

Message

सत्यमेव जयते

Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability dependents on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan'* is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.

15.4.13.

(Jitendra Singh)

Message

सत्यमेव जयते

Dr. Purushottam Agarwal
Principal Secretary to Government,
Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of 'Hydrogeological Atlas of Rajasthan' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword

European Union
State Partnership Programme

Arno Schaefer
Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

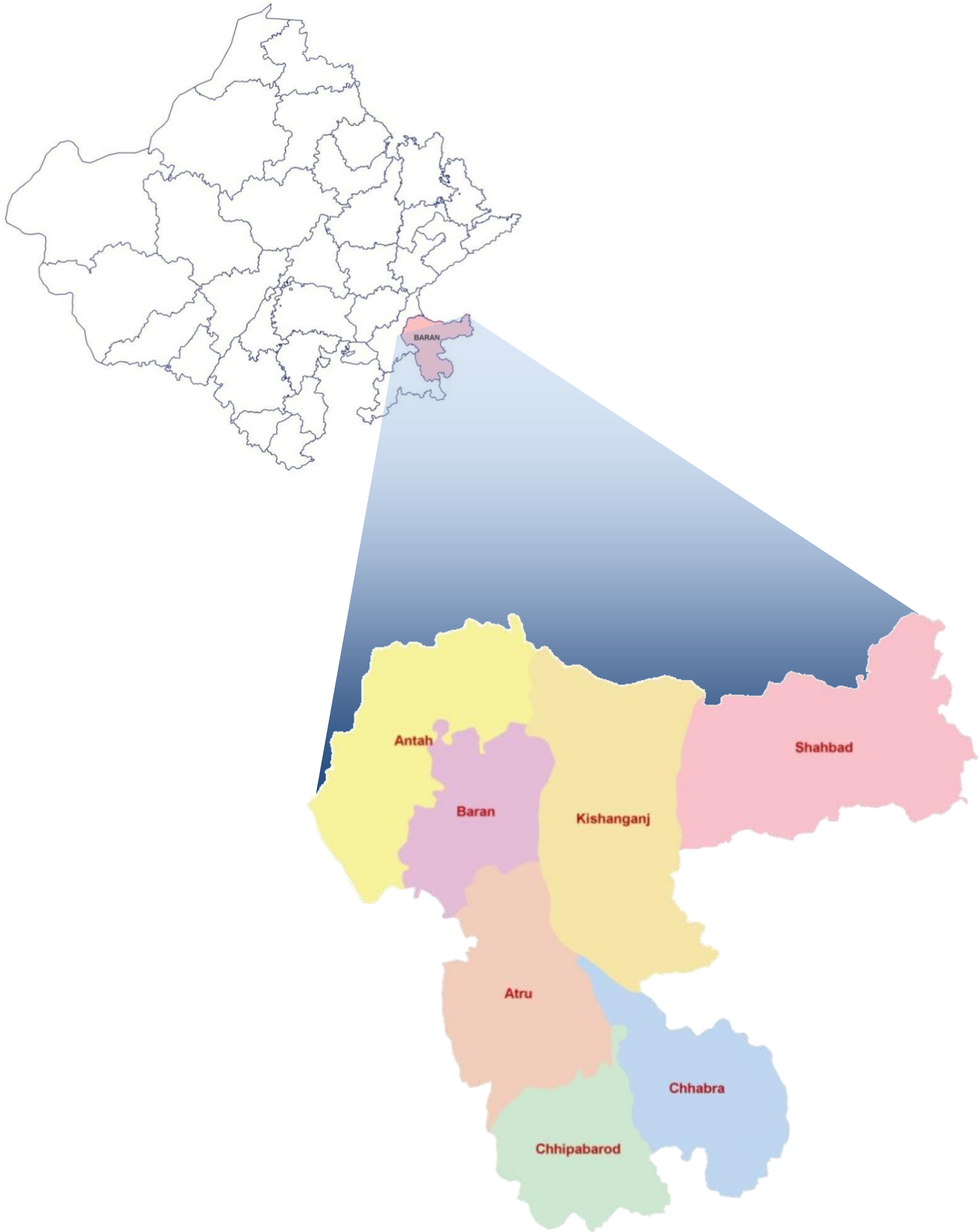
I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Baran District



Contents:

2013

Location:

Baran district is located in the southeastern part of Rajasthan. It is bounded in the east by state of Madhya Pradesh, southwest by Jhalawar district and northwest by Kota district. It stretches between 24^{o} 23' 35.85" to 25^{o} 26' 39.94" North latitude and 76^{o} 11' 34.16'' to 77^{o} 25' 56.74'' East longitude covering area of 6,994 sq kms. Major part of the district has a systematic drainage system, as whole district is part of 'Chambal River Basin'.

Administrative Set-up:

Baran district is administratively divided into seven Blocks. The following table summarizes the basic statistics of the district at block level.

S. No.	Block Name	Population (Based on 2001 census)	Area (sq km)	% of District Area	Total Number of Towns and Villages
1	Antah	1,97,385	1,033.3	14.8	158
2	Atru	1,32,944	946.4	13.5	139
3	Baran	1,81,807	638.8	9.1	103
4	Chhabra	1,22,268	798.9	11.4	193
5	Chhipabarod	1,43,885	684.3	9.8	181
6	Kishanganj	1,35,218	1,450.4	20.7	203
7	Shahbad	1,08,146	1,441.9	20.7	236
Total		**10,21,653**	**6,994.0**	**100.0**	**1,213**

Baran district has 1,213 towns and villages, of which seven are block headquarters as well.

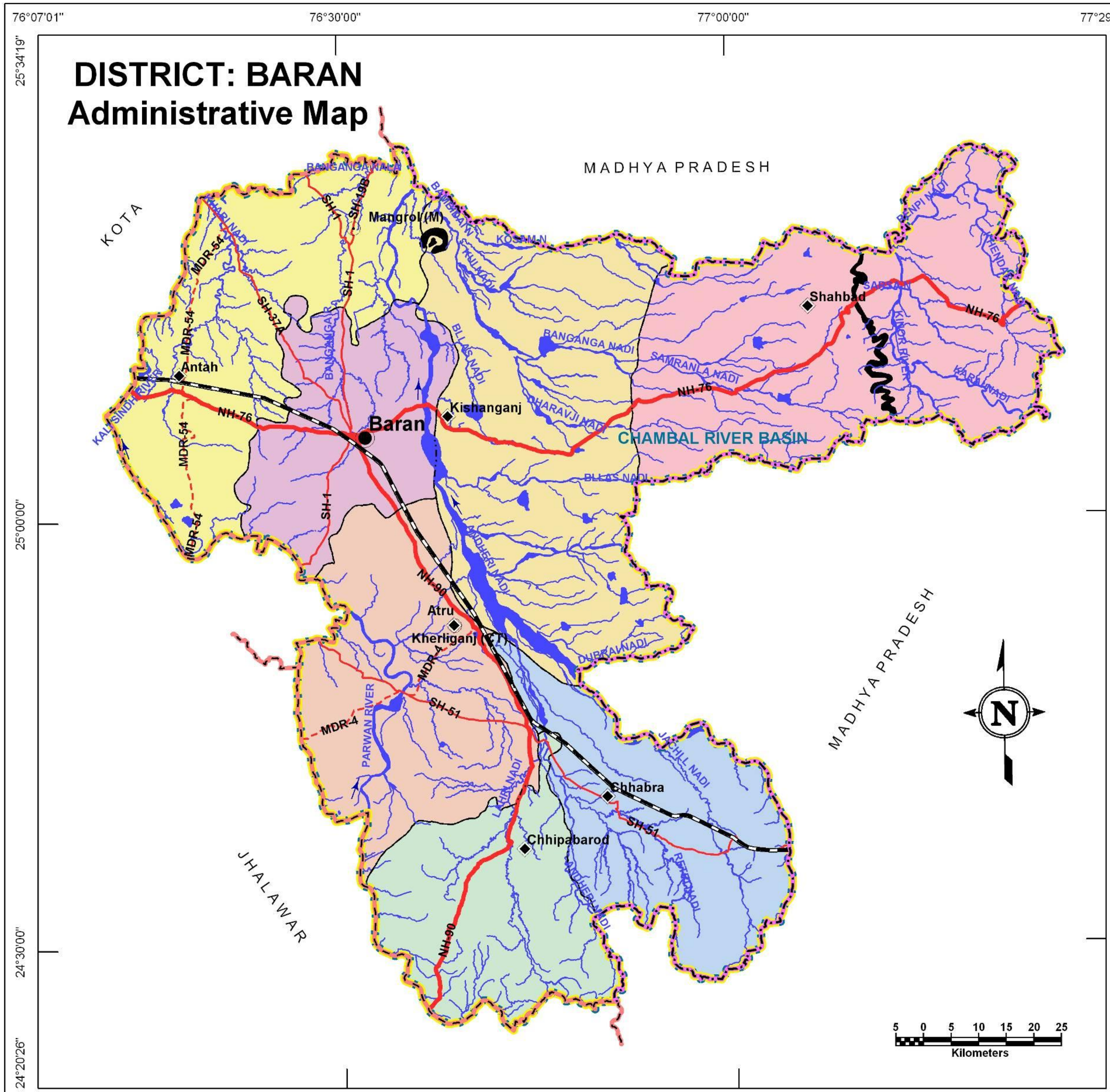
Climate:

The district has a sub-humid climate, moderately dry and receives fairly good rainfall in monsoon seasons. The winter season extends from November to February and summer season from March to mid of June. The period from mid of June to September is the monsoon season followed by the months October to mid of November constitutes the post monsoon or the retreating monsoon. The mean annual rainfall in the district is 838.7mm. January is the coldest month with the average daily maximum temperature of 24.3 ^{o}C and the average daily minimum temperature in the range of 8-10 ^{o}C.

Ground Water Department, Rajasthan

European Union
State Partnership Programme

PLATE - I



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

Basin Boundary

Roads:

- National Highway
- State Highway
- Major District Road

Railway:

- Broad Gauge

Water Bodies:

- River / Streams
- Ponds / Reservoirs

Hills

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TOPOGRAPHY

The topography of Baran district is in general, undulating in in the western part, but hilly in the eastern and southern parts constituting predominant landforms visible in Shahbad, Chhabra and Chhipabarod blocks. The general topographic elevation in the district is between 250 m to 450 m above mean sea level. The narrow elongated hills rise up to 548.8 m elevation in eastern part of the district. Apart from the well-known Chambal River, the other rivers that flow through the districts are Andheri and Kali Sindh and their tributaries are Bambidai, Kui, Kosam, Khari, Bllas and Samranla. Elevation ranges from a minimum of 200 m above mean sea level in Antah block in the northwestern part of the district to a maximum of 548.8 m above mean sea level In Shahbad block in eastern part of the district.

Table: Block wise minimum and maximum elevation

S. No.	Block Name	Minimum Elevation (m amsl)	Maximum Elevation (m amsl)
1	Antah	200.0	288.3
2	Atru	252.2	429.5
3	Baran	207.7	293.5
4	Chhabra	281.0	488.9
5	Chhipabarod	290.5	471.5
6	Kishanganj	212.2	442.9
7	Shahbad	283.9	548.8

RAINFALL

The rainfall is very scanty and erratic. The general distribution of rainfall across can be visualized from isohyets presented in the Plate – III where most of the district received rainfall in the range of 600-700 mm in year 2010. The annual average rainfall was 621.2 mm based on the data of available blocks. Maximum Annual rainfall was noticed in Atru block (724.4 mm) whereas minimum was in Antah block (562.0 mm). The highest average annual rainfall noticed in Chhipabarod block.

Table: Block wise annual rainfall statistics (derived from year 2010 meteorological station data)

Block Name	Minimum Annual Rainfall (mm)	Maximum Annual Rainfall (mm)	Average Annual Rainfall (mm)
Antah	445.2	697.8	614.9
Atru	584.6	724.4	667.9
Baran	512.8	639.8	599.5
Chhabra	557.1	685.8	590.6
Chhipabarod	616.3	705.5	675.2
Kishanganj	578.0	669.1	616.0
Shahbad	568.3	612.0	584.0

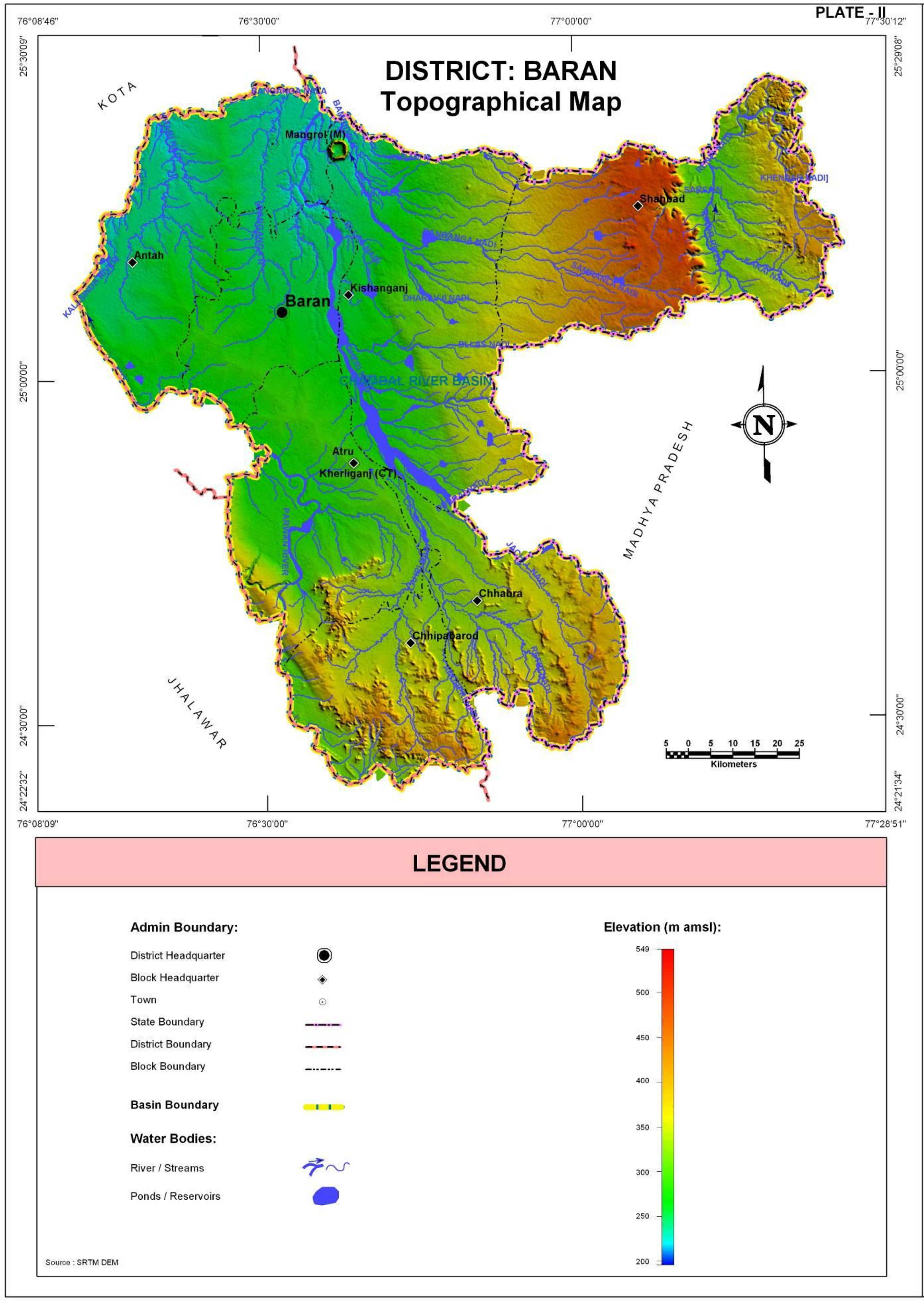


PLATE - II
DISTRICT: BARAN
Topographical Map
KOTA
MADHYA PRADESH
JHALAWAR
Mangrol (M)
Shahbad
Antah
Baran
Kishanganj
CHAMBAL RIVER BASIN
Atru
Kherliganj (CT)
Chhabra
Chhipabarod
N
Kilometers
LEGEND
Admin Boundary:
District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary
Basin Boundary
Water Bodies:
River / Streams
Ponds / Reservoirs
Elevation (m amsl):
549
500
450
400
350
300
250
200
Source : SRTM DEM

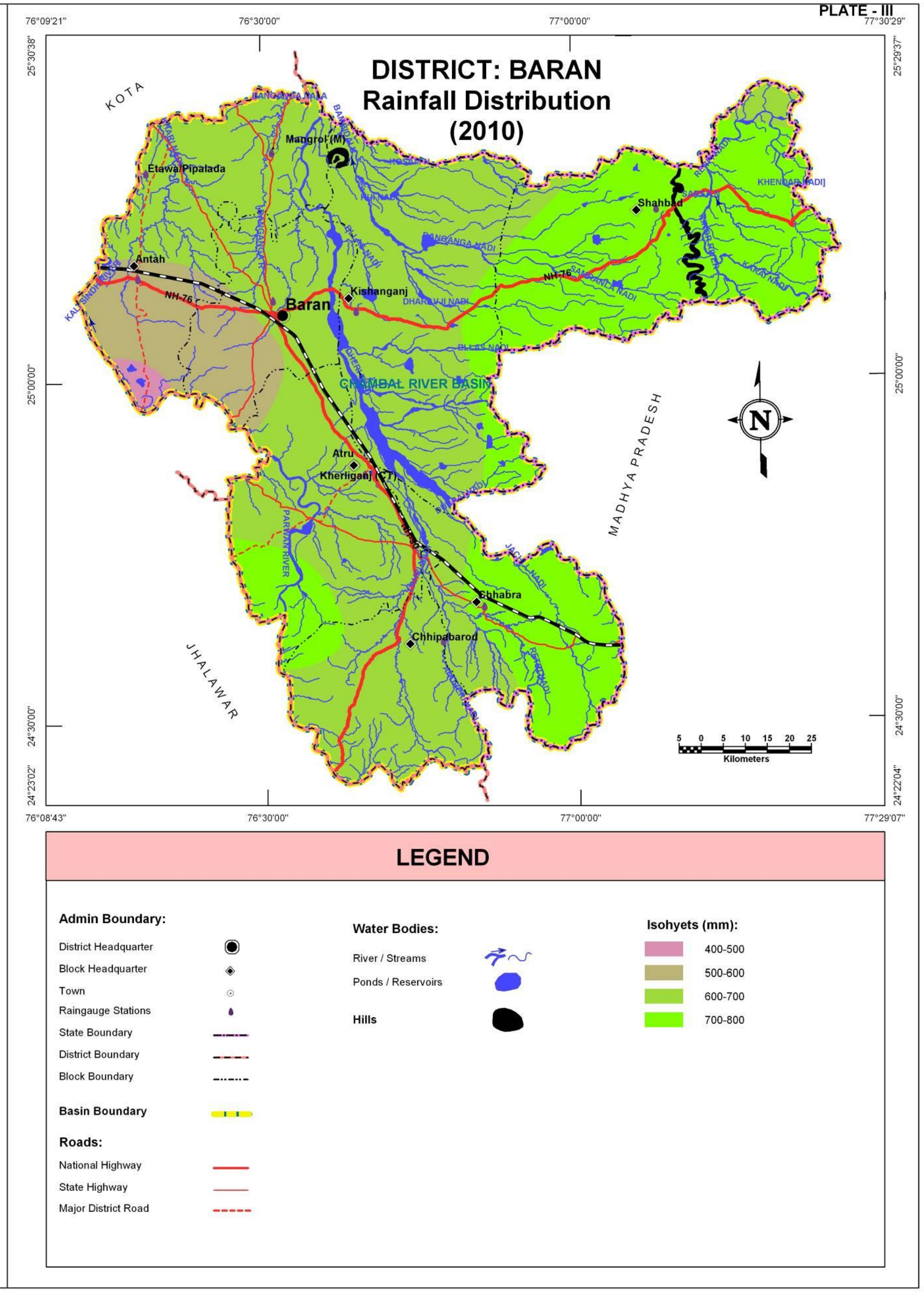


PLATE - III
DISTRICT: BARAN
Rainfall Distribution
(2010)
KOTA
MADHYA PRADESH
JHALAWAR
Mangrol (M)
Etawa Pipalada
Shahbad
Antah
Baran
Kishanganj
NH-76
CHAMBAL RIVER BASIN
Atru
Kherliganj (CT)
Chhabra
Chhipabarod
N
Kilometers
LEGEND
Admin Boundary:
District Headquarter
Block Headquarter
Town
Raingauge Stations
State Boundary
District Boundary
Block Boundary
Basin Boundary
Roads:
National Highway
State Highway
Major District Road
Water Bodies:
River / Streams
Ponds / Reservoirs
Hills
Isohyets (mm):
400-500
500-600
600-700
700-800

GEOLOGY

Most of the part of Baran district is occupied by rocks of Vindhyan Super Group. On the basis of different rock-units the Vindhyans of the area have been divided in to Bhander, Rewa and Khorip Groups comprising Shale, Limestone and Sandstones. The Khorip Group is separated from Rewa Group by conglomerate horizon. The Southern part of Chhabra and Chhipabarod block is occupied by Deccan trap basalt. Bhander Group occupies most part of the district.

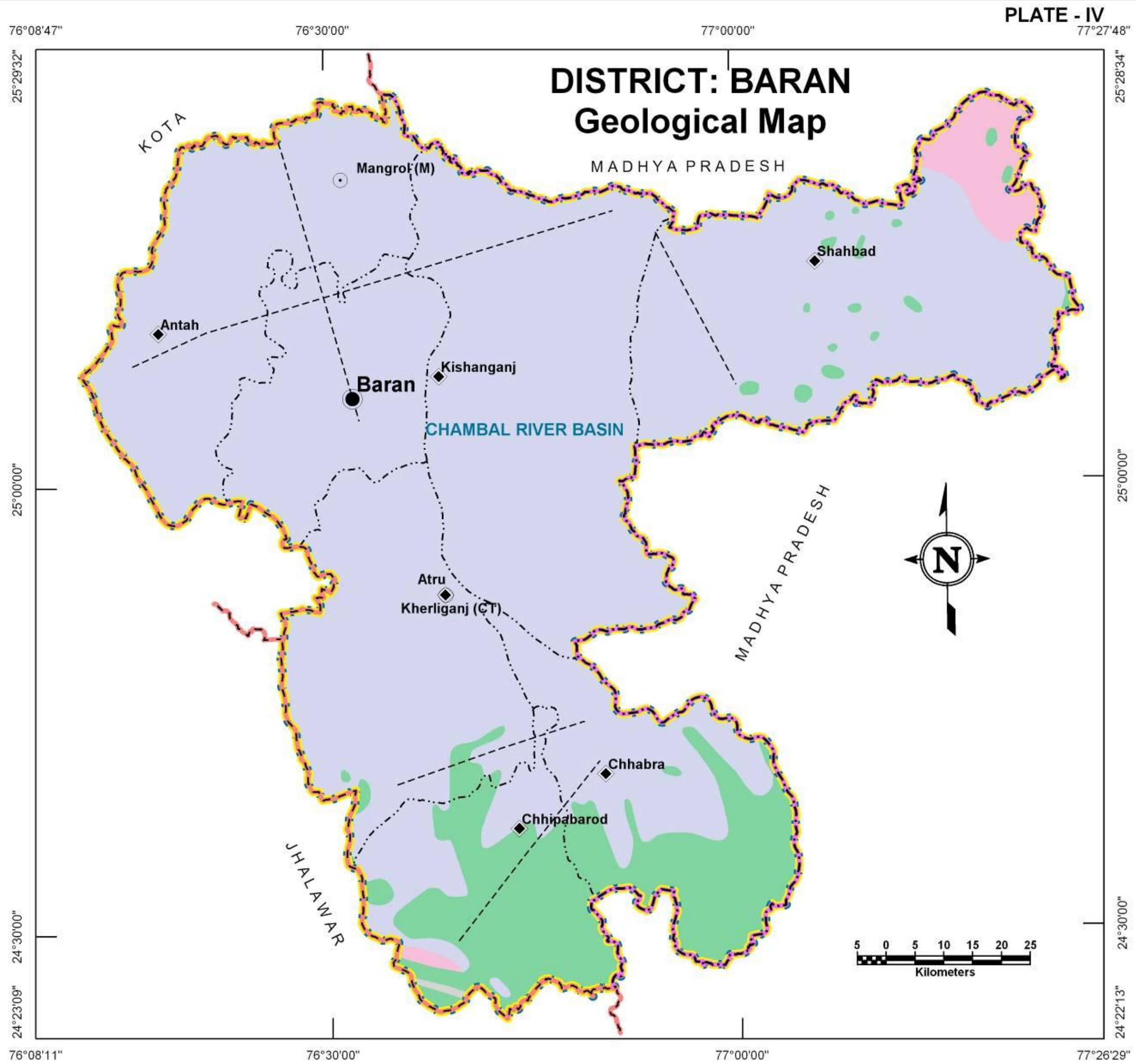
Super Group	Group	Formation
	Recent	Alluvium (Sand, Silt, Clay)
	Deccan trap	Basalt
--X--------X--------X--------X---Unconformity---X--------X--------X-------X--		
Vindhyan	Bhander	Shale, Limestone, Sandstone
	Rewa	Shale
--X--------X--------X--------X---Unconformity---X--------X--------X-------X--		
Lower Vindhyan	Khorip	Shale, Limestone, Sandstone

GEOMORPHOLOGY

Table: Geomorphologic units, their description and distribution

Origin	Landform Unit	Description
Denudational	Buried Pediment	Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.
	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.
Fluvial	Alluvial Plain	Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.
	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
	Ravine	Small, narrow, deep, depression, smaller than gorges, larger than gulley, usually carved by running water.
Structural	Dissected Plateau	Plateau, criss-crossed by fractures forming deep valleys.
Hills	Denudational, Structural Hill, Linear Ridge	Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike.

PLATE - IV



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- **Basin Boundary**

Structural Features:

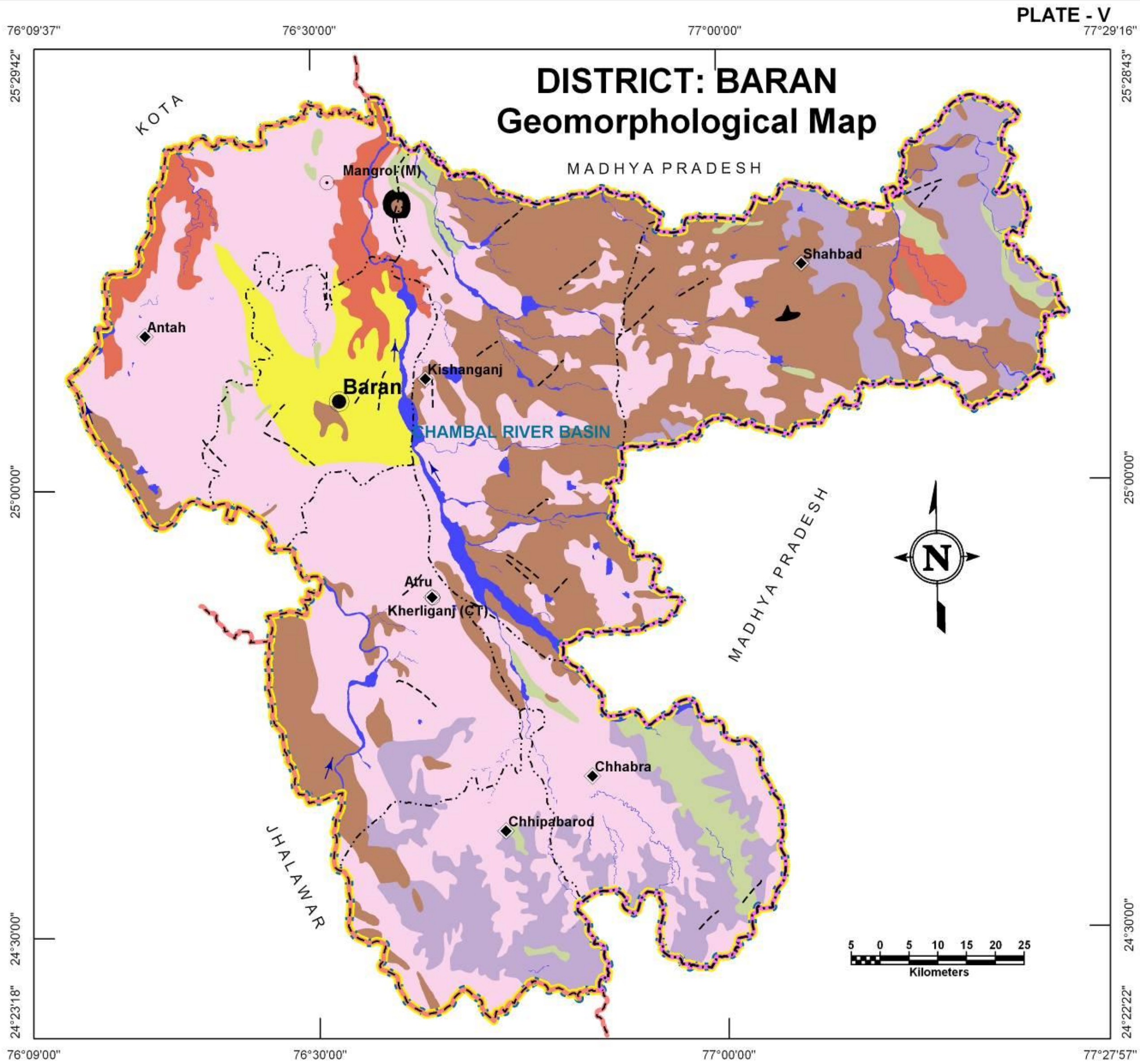
- Lineament

Geology:

- Bhander Group
- Rewa Group
- Khorip Group
- Deccan Traps

Source: District Resource Map of Rajasthan - GSI

PLATE - V



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- **Basin Boundary**

Water Bodies:

- River/Ponds/Reservoirs

Structural Features:

- Fault/Fractures/Lineament

Hills:

- Structural/Denudational/Linear Ridge

Landform Units:

Fluvial Origin:

- Alluvial Plain
- Valley Fill
- Ravine

Denudational Origin:

- Burried Pediment
- Pediment

Structural Origin:

- Dissected Plateau

Source: Ground Water Atlas of Rajasthan - SRSAC & GWD, Rajasthan

AQUIFERS

Aquifers in Baran district are formed in Sandstone, Limestone, Basalt, Shale and Younger alluvium. Weathered and fractured parts of the massive rocks along with primary and secondary openings in Sandstones and Limestone respectively, contribute to aquifer formation whereas the sandy, gravelly and other granular parts of alluvium constitute aquifers. Vindhyan sandstones are the most prevalent aquifer types with more than 52% spatial coverage and present prominently in the central part of the district. The Basalt aquifers occupy about 16% of the district area and present in southern part of the district. Shales and Limestones constitute aquifers in the eastern and western parts of the district respectively. Alluvium constitutes small patch of sandy aquifers around Baran contributing to less than 5% of aquifer area to the district.

Table: aquifer potential zones their area and their description

Aquifer in Potential Zone	Area (sq km)	% age of district	Description of the unit/Occurrence
Younger Alluvium	327.1	4.7	It is largely constituted of Aeolian and Fluvial sand, silt, clay, gravel and pebbles in varying proportions.
Limestone	1204.3	17.2	In general, it is fine to medium grained, grey, red yellowish, pink or buff in colour.
Sandstone	3670.0	52.5	Fine to medium grained, red colour and compact and at places.
Shale	622.8	8.9	Grey, light green and purple in colour and mostly splintery in nature.
Basalt	1,141.7	16.3	Dark grey, olive green and green colour, compact, vesicular, amygdaloidal and weathered.
Hills	28.1	0.4	
Total	**6,994.0**	**100.0**	

STAGE OF GROUND WATER DEVELOPMENT

The different blocks of the district fall into three categories of ground water development stages. Shahbad, the easternmost block falls into 'safe' category whereas further west the Kishangarh block is under 'semi-critical' category. The rest of the blocks constituting a strip along the western part of the district, fall under over-exploited category indicating the areas where steps need to be taken to reduce stress on ground water due to exploitation.

Categorization on the basis of stage of development of ground water	Block Name
Safe	Shahbad
Semi- Critical	Kishanganj
Over Exploited	Antah, Baran, Atru, Chhipabarod, Chhabra.

Basis for categorization: Ground water development <= 70% - Safe; >70%; >90% - Semi critical and >100% - Over-Exploited.

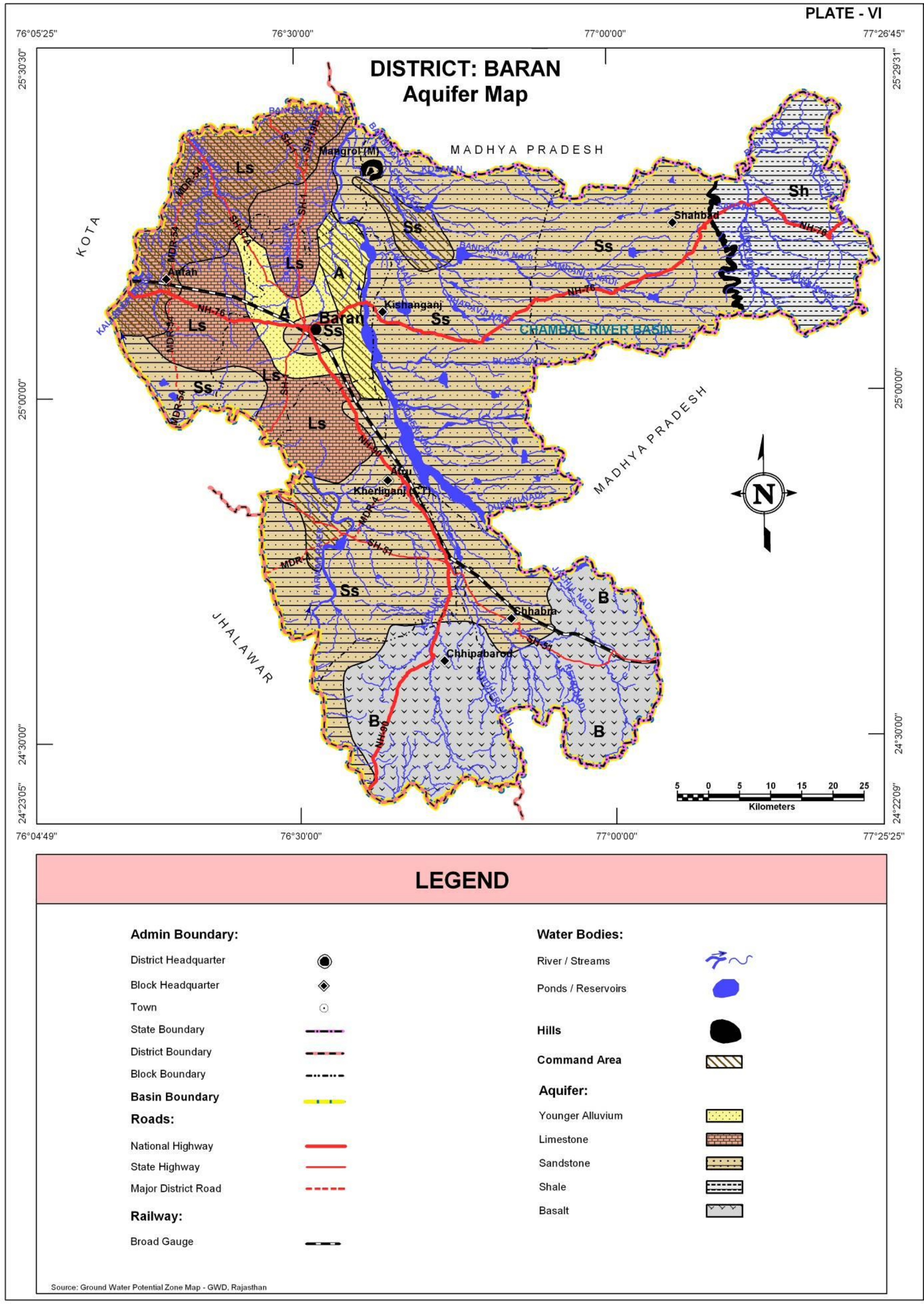


PLATE - VI
DISTRICT: BARAN
Aquifer Map
MADHYA PRADESH
KOTA
JHALAWAR
CHAMBAL RIVER BASIN
Mangrol (M)
Antah
Baran
Kishanganj
Shahbad
Atru
Kherliganj (CT)
Chhabra
Chhipabarod
Ls
Ss
Sh
A
B
Kilometers
LEGEND
Admin Boundary:
District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary
Basin Boundary
Roads:
National Highway
State Highway
Major District Road
Railway:
Broad Gauge
Water Bodies:
River / Streams
Ponds / Reservoirs
Hills
Command Area
Aquifer:
Younger Alluvium
Limestone
Sandstone
Shale
Basalt
Source: Ground Water Potential Zone Map - GWD, Rajasthan

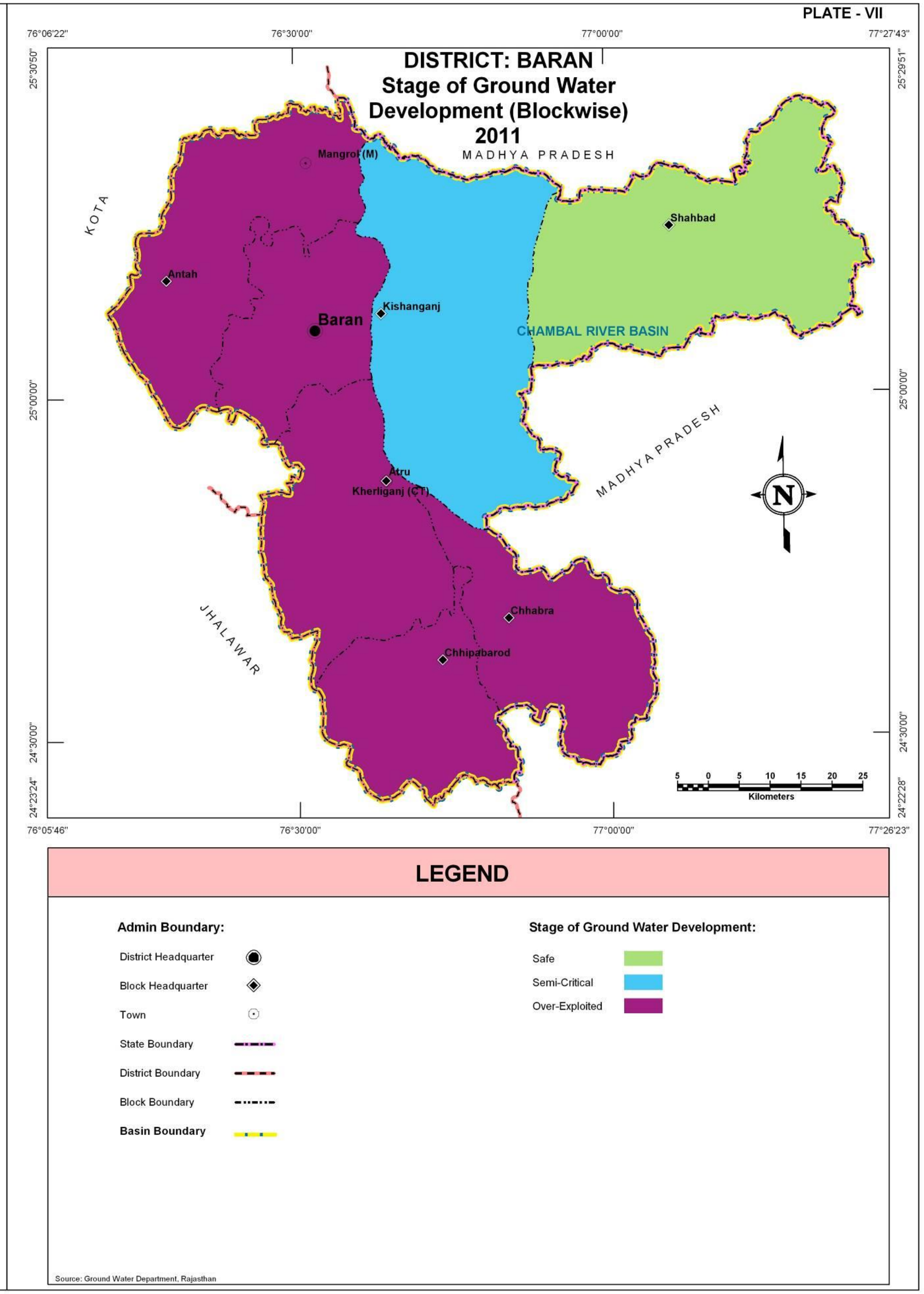


PLATE - VII
DISTRICT: BARAN
Stage of Ground Water
Development (Blockwise)
2011
MADHYA PRADESH
KOTA
JHALAWAR
CHAMBAL RIVER BASIN
Mangrol (M)
Antah
Baran
Kishanganj
Shahbad
Atru
Kherliganj (CT)
Chhabra
Chhipabarod
Kilometers
LEGEND
Admin Boundary:
District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary
Basin Boundary
Stage of Ground Water Development:
Safe
Semi-Critical
Over-Exploited
Source: Ground Water Department, Rajasthan

LOCATION OF EXPLORATORY AND GROUND WATER MONITORING WELLS DISTRICT – BARAN

The district has a well distributed network of large number of exploratory wells (47) and ground water monitoring stations (170) in the district owned by RGWD (43 and 154 respectively) and CGWB (4 and 16 respectively). The exploratory wells have formed the basis for delineation of subsurface aquifer distribution scenario in three dimensions. Benchmarking and optimization studies suggest that ground water levels are being well monitored through existing network but for better monitoring of ground water quality five wells need to be added to existing network in Baran block.

Table: Block wise count of wells (existing and recommended)

Block Name	Exploratory Wells			Ground Water Monitoring Stations			Recommended additional wells for optimization of monitoring network	
	CGWB	RGWD	Total	CGWB	RGWD	Total	Water Level	Water Quality
Antah	1	9	**10**	2	31	**33**	-	-
Atru	1	7	**8**	3	25	**28**	-	-
Baran	1	6	**7**	2	24	**26**	-	5
Chhabra	1	5	**6**	1	15	**16**	-	-
Chhipabarod	-	5	**5**	3	19	**22**	-	-
Kishanganj	-	7	**7**	2	19	**21**	-	-
Shahbad	-	4	**4**	3	21	**24**	-	-
Total	**4**	**43**	**47**	**16**	**154**	**170**	**0**	**5**

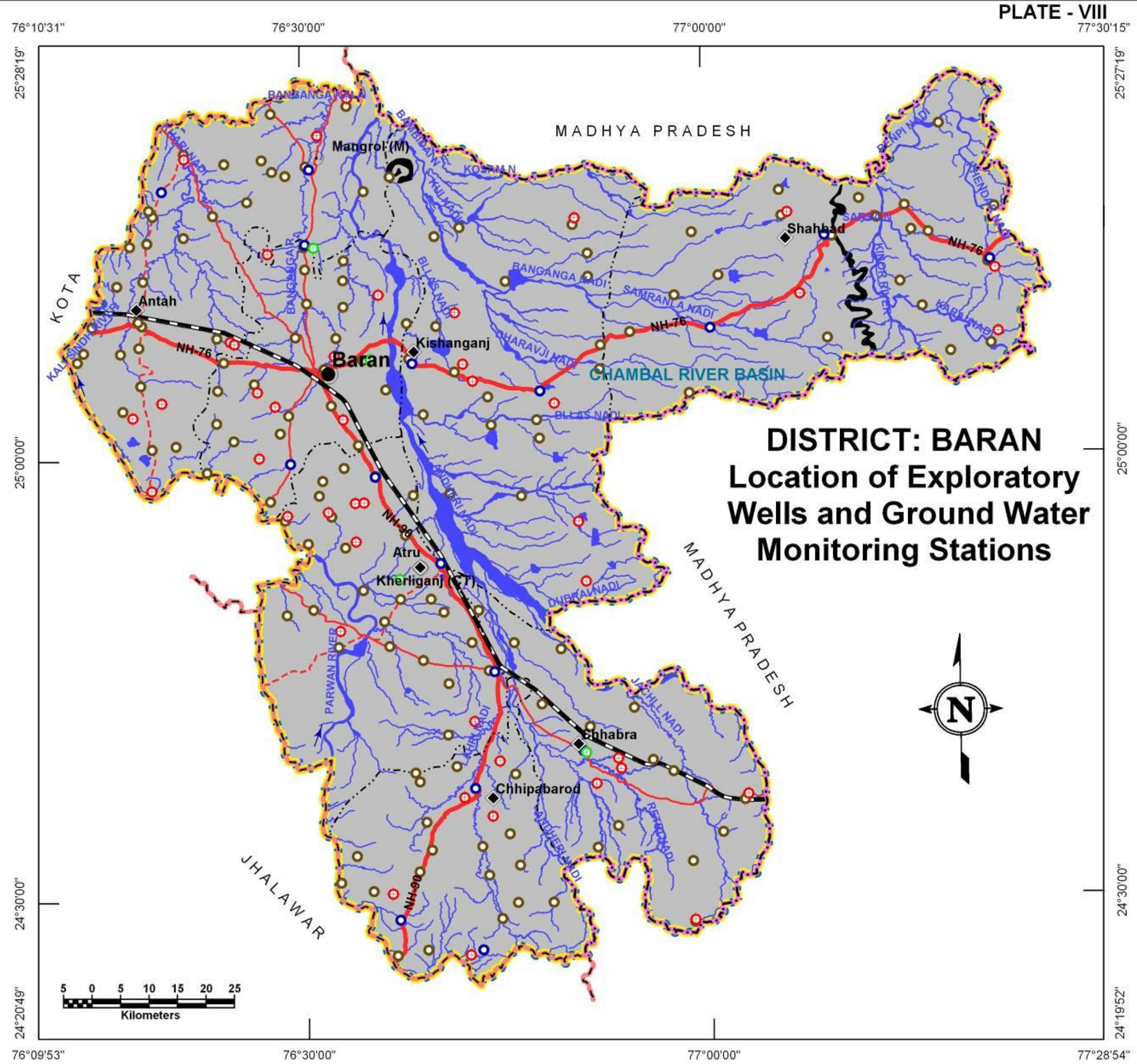
DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

The district shows variation in depth to ground water levels from less than 10m bgl to around 60m bgl. Ground water occurs at shallow depths of less than 10m below ground level in almost half of the district area. 10 – 20m deep water levels are also found in about 38% of the district. Combined together, the areas having less than 20m depth to water level occupy almost 83% of the district. The deeper water levels are seen distributed in the western part of Chambal river in Atru, Baran, Chhipabarod blocks and small area in Shahbad towards eastern part of the district.

Depth to water level (m bgl)	Block wise area coverage (sq km) *							Total Area (sq km)
	Antah	Atru	Baran	Chhabra	Chhipabarod	Kishanganj	Shahbad	
<10	492.1	14.4	75.1	415.0	290.9	674.8	1,148.6	**3,110.9**
10-20	498.8	464.3	234.1	269.1	316.3	653.9	235.7	**2,672.2**
20-30	41.8	242.7	153.6	93.7	39.9	116.0	19.6	**707.3**
30-40	-	134.8	136.5	21.1	32.1	-	9.5	**334.0**
40-50	-	73.3	32.5	-	5.0	-	5.2	**116.0**
>50	-	16.7	7.0	-	-	-	1.8	**25.5**
Total	**1,032.7**	**946.2**	**638.8**	**798.9**	**684.2**	**1,444.7**	**1,420.4**	**6,965.9**

* The area covered in the derived maps is less than the total district area since the hills have been excluded from interpolation/contouring.

PLATE - VIII



LEGEND

Admin Boundary:
- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- **Basin Boundary**

Roads:
- National Highway
- State Highway
- Major District Road

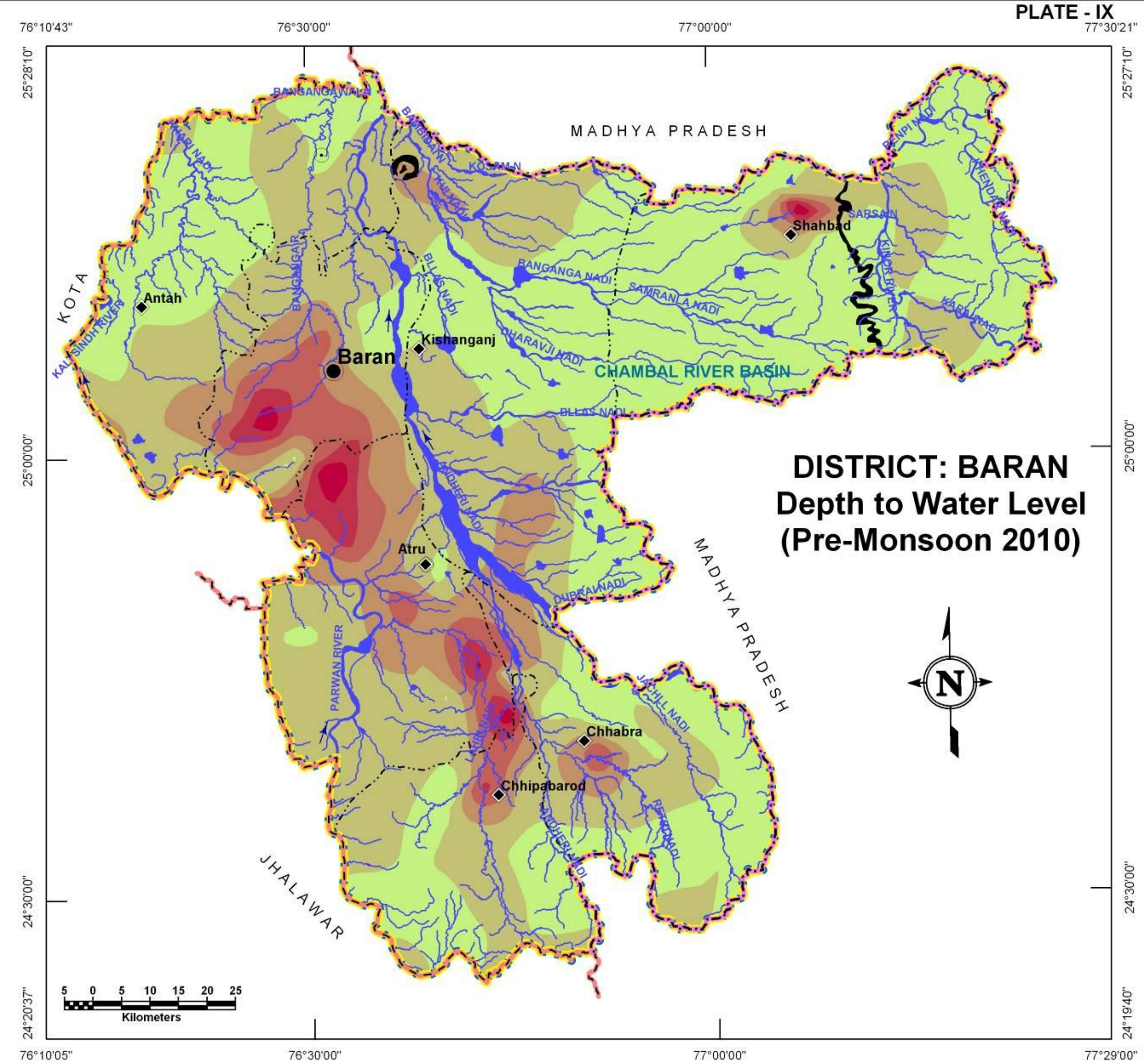
Railway
- Broad Gauge

Water Bodies:
- River / Streams
- Ponds / Reservoirs
- **Hills**

Exploratory Wells:
- CGWB
- RGWD

Ground Water Monitoring Stations:
- CGWB
- RGWD

PLATE - IX



LEGEND

Admin Boundary:
- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- **Basin Boundary**

Water Bodies:
- River / Streams
- Ponds / Reservoirs
- **Hills**

Depth to Water Level (m bgl):
- < 10
- 10 - 20
- 20 - 30
- 30 - 40
- 40 - 50
- > 50

WATER TABLE ELEVATION (PRE MONSOON – 2010) DISTRICT – BARAN

Water table contour map reveals the general flow direction of ground water in the district which is indicated to be in a broadly east/southeast to northwestern direction following topography as well as the direction of rivers. Variation in water table elevation seems to be high in the district as the maximum water table seen is >440m amsl in northeastern part in Shahbad block and minimum is in Antah block (<180m amsl) in the northwestern part of the district. The major part of the district shows water table in the range of 220m amsl to 380m amsl.

Block Name	Block wise area coverage (sq km) per water table elevation (m amsl) range														Total Area (sq km)
	< 180	180 - 200	200 - 220	220 - 240	240 - 260	260 - 280	280 - 300	300 - 320	320 - 340	340 - 360	360 - 380	380 - 400	400 - 440	> 440	
Antah	3.2	11.9	52.2	459.1	482.4	23.9	-	-	-	-	-	-	-	-	**1,032.7**
Atru	-	-	-	-	203.9	268.1	185.0	166.2	104.1	18.8	0.1	-	-	-	**946.2**
Baran	-	-	-	139.2	467.0	32.6	-	-	-	-	-	-	-	-	**638.8**
Chhabra	-	-	-	-	-	22.4	30.5	33.3	147.2	133.2	198.2	161.5	72.6	-	**798.9**
Chhipabarod	-	-	-	-	-	-	0.8	98.8	203.4	191.1	186.5	3.6	-	-	**684.2**
Kishanganj	-	-	5.3	61.6	155.5	257.3	253.3	257.5	261.0	175.2	18.0	-	-	-	**1,444.7**
Shahbad	-	-	-	-	-	-	38.7	126.8	259.9	212.1	190.1	154.7	406.3	31.8	**1,420.4**
Total	**3.2**	**11.9**	**57.5**	**659.9**	**1,308.8**	**604.3**	**508.3**	**682.6**	**975.6**	**730.4**	**592.9**	**319.8**	**478.9**	**31.8**	**6,965.9**

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

A 2m contour interval adopted to visualize the ground water level fluctuation reveals a fall of 6 m in one area while rise in other areas reaching upto more than 14m, as seen in Plate XI. The –ve fluctuation areas (indicated by red colored regions) are the areas where overexploitation is taking place and even after monsoon recharge water level has not risen and has actually gone down with respect to pre-monsoon levels. Such large ground water depletion areas are located in the northern and southern part of the district. Rest of the district has shown a general to significant rise in ground water level in the post monsoon season with respect to pre monsoon region. Maximum rise of more than 14m is noticed at northern part of Chhabra.

Block Name	Blockwise area coverage (sq km) per Water level fluctuation range (m)												Total Area (sq km)
	<-6	-6 - -4	-4 - -2	-2 - 0	0 - 2	2 - 4	4 - 6	6 - 8	8 - 10	10 - 12	12 - 14	>14	
Antah	3.5	7.8	30.1	308.8	440.5	149.8	65.4	26.5	0.3	-	-	-	**1,032.7**
Atru	-	-	44.1	149.1	254.7	212.4	141.1	84.3	39.6	18.7	2.2	-	**946.2**
Baran	-	0.7	21.8	185.1	224.2	134.5	57.3	10.6	2.0	2.6	-	-	**638.8**
Chhabra	-	-	-	164.3	319.2	162.9	81.0	37.8	20.5	9.1	3.7	0.4	**798.9**
Chhipabarod	-	0.9	18.2	160.7	249.3	222.3	28.0	4.0	0.8	-	-	-	**684.2**
Kishanganj	-	-	-	61.9	691.9	469.0	180.0	41.9	-	-	-	-	**1,444.7**
Shahbad	-	-	-	13.6	451.9	771.5	174.5	8.9	-	-	-	-	**1,420.4**
Total	**3.5**	**9.4**	**114.2**	**1,043.5**	**2,631.7**	**2,122.4**	**727.3**	**214.0**	**63.2**	**30.4**	**5.9**	**0.4**	**6,965.9**

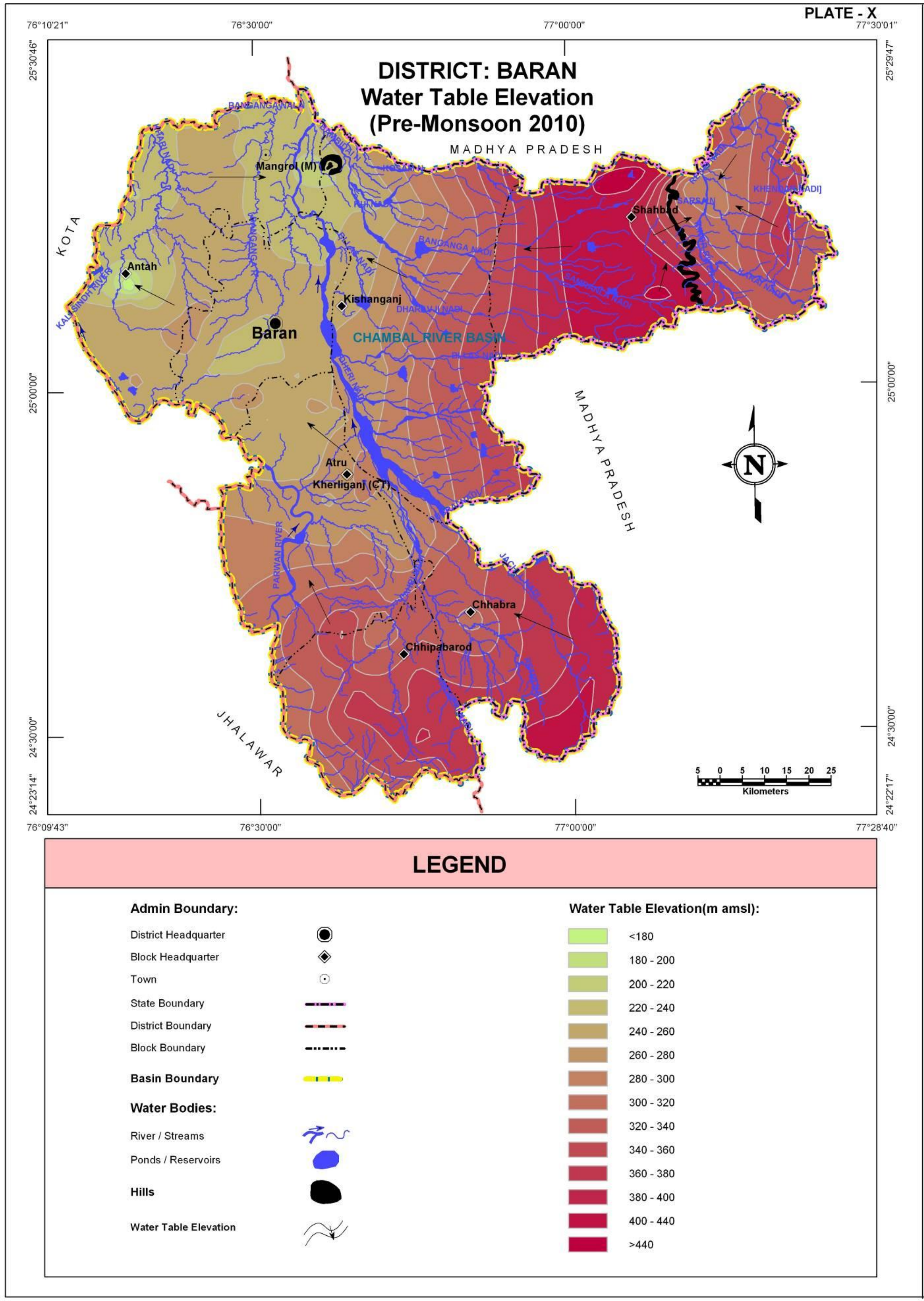


PLATE - X
DISTRICT: BARAN
Water Table Elevation
(Pre-Monsoon 2010)
MADHYA PRADESH
KOTA
JHALAWAR
Mangrol (M)
Shahbad
Antah
Kishanganj
Baran
CHAMBAL RIVER BASIN
Atru
Kherliganj (CT)
Chhabra
Chhipabarod
Kilometers
LEGEND
Admin Boundary:
District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary
Basin Boundary
Water Bodies:
River / Streams
Ponds / Reservoirs
Hills
Water Table Elevation
Water Table Elevation(m amsl):
<180
180 - 200
200 - 220
220 - 240
240 - 260
260 - 280
280 - 300
300 - 320
320 - 340
340 - 360
360 - 380
380 - 400
400 - 440
>440

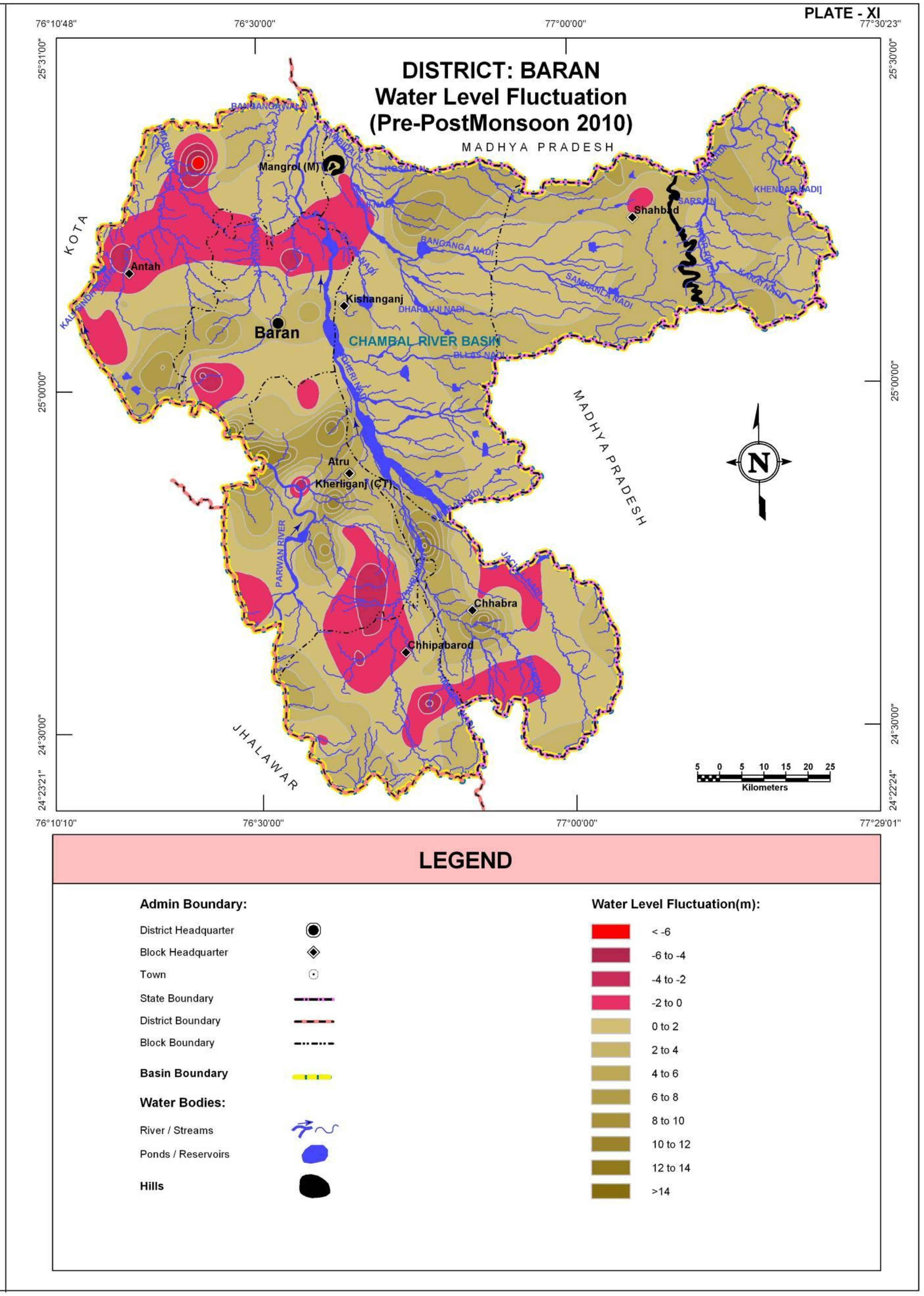


PLATE - XI
DISTRICT: BARAN
Water Level Fluctuation
(Pre-PostMonsoon 2010)
MADHYA PRADESH
KOTA
JHALAWAR
Mangrol (M)
Shahbad
Antah
Kishanganj
Baran
CHAMBAL RIVER BASIN
Atru
Kherliganj (CT)
Chhabra
Chhipabarod
Kilometers
LEGEND
Admin Boundary:
District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary
Basin Boundary
Water Bodies:
River / Streams
Ponds / Reservoirs
Hills
Water Level Fluctuation(m):
< -6
-6 to -4
-4 to -2
-2 to 0
0 to 2
2 to 4
4 to 6
6 to 8
8 to 10
10 to 12
12 to 14
>14

GROUND WATER ELECTRICAL CONDUCTIVITY DISTRIBUTION — DISTRICT – BARAN

The Electrical conductivity (at 25^{o}C) distribution map is presented in plate XII. The areas with low EC values in ground water (<2000 µS/cm) are shown in yellow color and occupies approximately 93% of the district area indicating that, by and large the ground water in this region is suitable for domestic purpose. The area with moderately high EC values (2000-4000 µS/cm) is shown in green color occupies approximately 7% of the district area, largely around Mangrol and Baran. In this district the areas with high EC values (>4000 µS/cm) occupy negligibly small area as shown in red color in parts of Baran, Antah and Atru blocks.

Table: Block wise area of Electrical conductivity distribution

Electrical Conductivity Ranges (µS/cm at 25^{o}C) (Ave. of years 2005-09)	Block wise area coverage (sq km)														Total Area (sq km)
	Antah		Atru		Baran		Chhabra		Chhipabarod		Kishanganj		Shahbad		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
<2000	839.7	81.3	933.1	98.6	456.1	71.4	795.8	99.6	684.2	100.0	1,391.6	96.3	1,357.9	95.6	**6,458.4**
2000-4000	178.5	17.3	11.1	1.2	170.0	26.6	3.1	0.4	-	-	53.1	3.7	62.5	4.4	**478.3**
>4000	14.5	1.4	2.0	0.2	12.7	2.0	-	-	-	-	-	-	-	-	**29.2**
Total	**1,032.7**	**100.0**	**946.2**	**100.0**	**638.8**	**100.0**	**798.9**	**100.0**	**684.2**	**100.0**	**1,444.7**	**100.0**	**1,420.4**	**100.0**	**6,965.9**

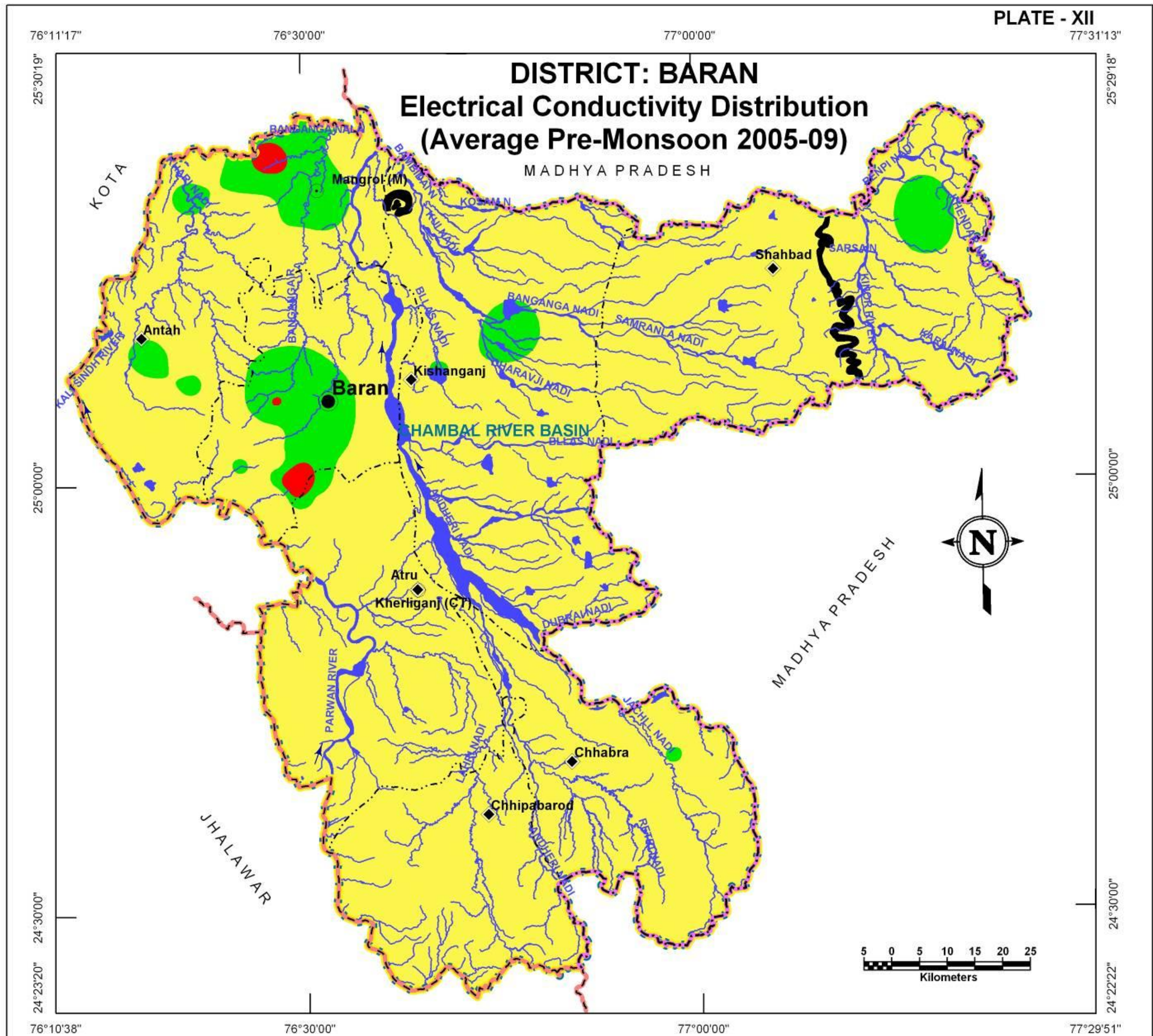
GROUND WATER CHLORIDE DISTRIBUTION

Chloride distribution map is presented in Plate XIII. The yellow colored regions in map are such areas where chloride concentration is low (<250 mg/l) and such areas occupy approximately 92% of the district area which is suitable for domestic purposes. The areas with moderately high chloride concentration (250-1000 mg/l) are shown in green color and occupy approximately 8%of the district area, largely around Mangrol and Baran. In this district the area with high chloride concentration (>1000 mg/l) occupies negligibly small areas seen in baran block only.

Table: Block wise area of Chloride distribution

Chloride Concentration Range (mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)														Total Area (sq km)
	Antah		Atru		Baran		Chhabra		Chhipabarod		Kishanganj		Shahbad		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
<250	900.1	87.0	934.6	99.0	435.9	68.0	798.9	100.0	684.2	100.0	1,351.4	94.0	1,242.0	87.0	**6,347.1**
250-1000	132.6	13.0	11.6	1.0	198.8	31.0	-	-	-	-	93.3	6.0	178.4	13.0	**614.7**
>1000	-	-	-	-	4.1	1.0	-	-	-	-	-	-	-	-	**4.1**
Total	**1,032.7**	**100.0**	**946.2**	**100.0**	**638.8**	**100.0**	**798.9**	**100.0**	**684.2**	**100.0**	**1,444.7**	**100.0**	**1,420.4**	**100.0**	**6,965.9**

PLATE - XII



LEGEND

Admin Boundary:

District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary

Basin Boundary

Water Bodies:

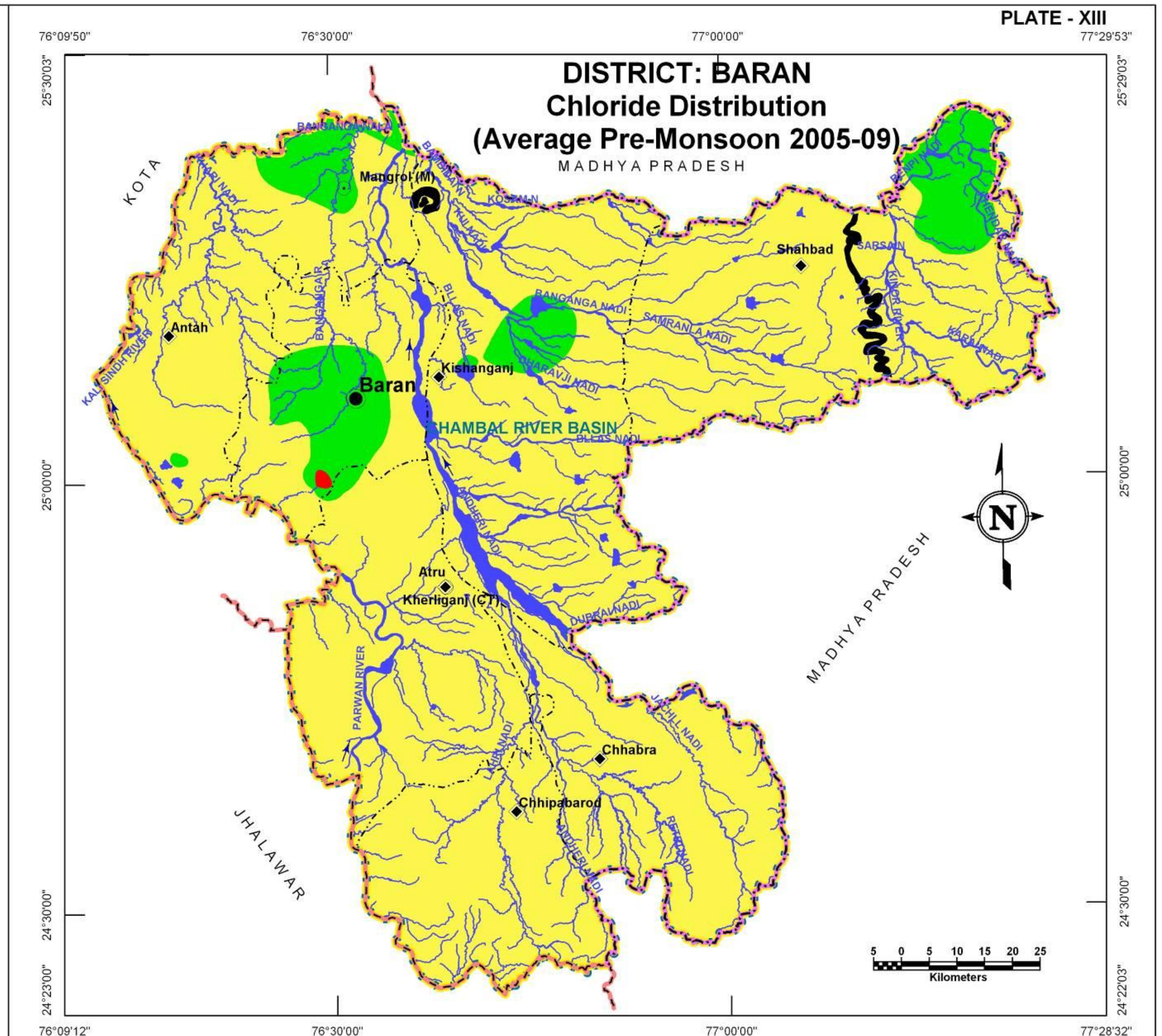
River / Streams
Ponds / Reservoirs

Hills

Electrical Conductivity (μS/cm at 25°C):

< 2000
2000-4000
>4000

PLATE - XIII



LEGEND

Admin Boundary:

District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary

Basin Boundary

Water Bodies:

River / Streams
Ponds / Reservoirs

Hills

Chloride Concentration (mg/l):

< 250
250 - 1000
> 1000

GROUND WATER FLUORIDE DISTRIBUTION DISTRICT – BARAN

The Fluoride concentration map is presented in Plate – XIV. The areas with low concentration (i.e.,>1.5 mg/l) are shown in yellow color and occupy approximately 97% of the total district area indicating that, by and large the ground water in this district is suitable for domestic purpose. The areas with moderately high concentration (1.5 – 3.0 mg/l) are shown in green color, largely northeastern part of the district. There is no area that had shown very high fluoride concentration in ground water.

Table: Block wise area of Fluoride distribution

Fluoride concentration Range (mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)														Total Area (sq km)
	Antah		Atru		Baran		Chhabra		Chhipabarod		Kishanganj		Shahbad		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
<1.5	1,032.7	100.0	946.2	100.0	638.8	100.0	798.9	100.0	684.2	100.0	1,444.7	100.0	1,180.0	83.1	6,725.5
1.5-3.0	-	-	-	-	-	-	-	-	-	-	-	-	240.4	16.9	240.4
Total	**1,032.7**	**100.0**	**946.2**	**100.0**	**638.8**	**100.0**	**798.9**	**100.0**	**684.2**	**100.0**	**1,444.7**	**100.0**	**1,420.4**	**100.0**	**6,965.9**

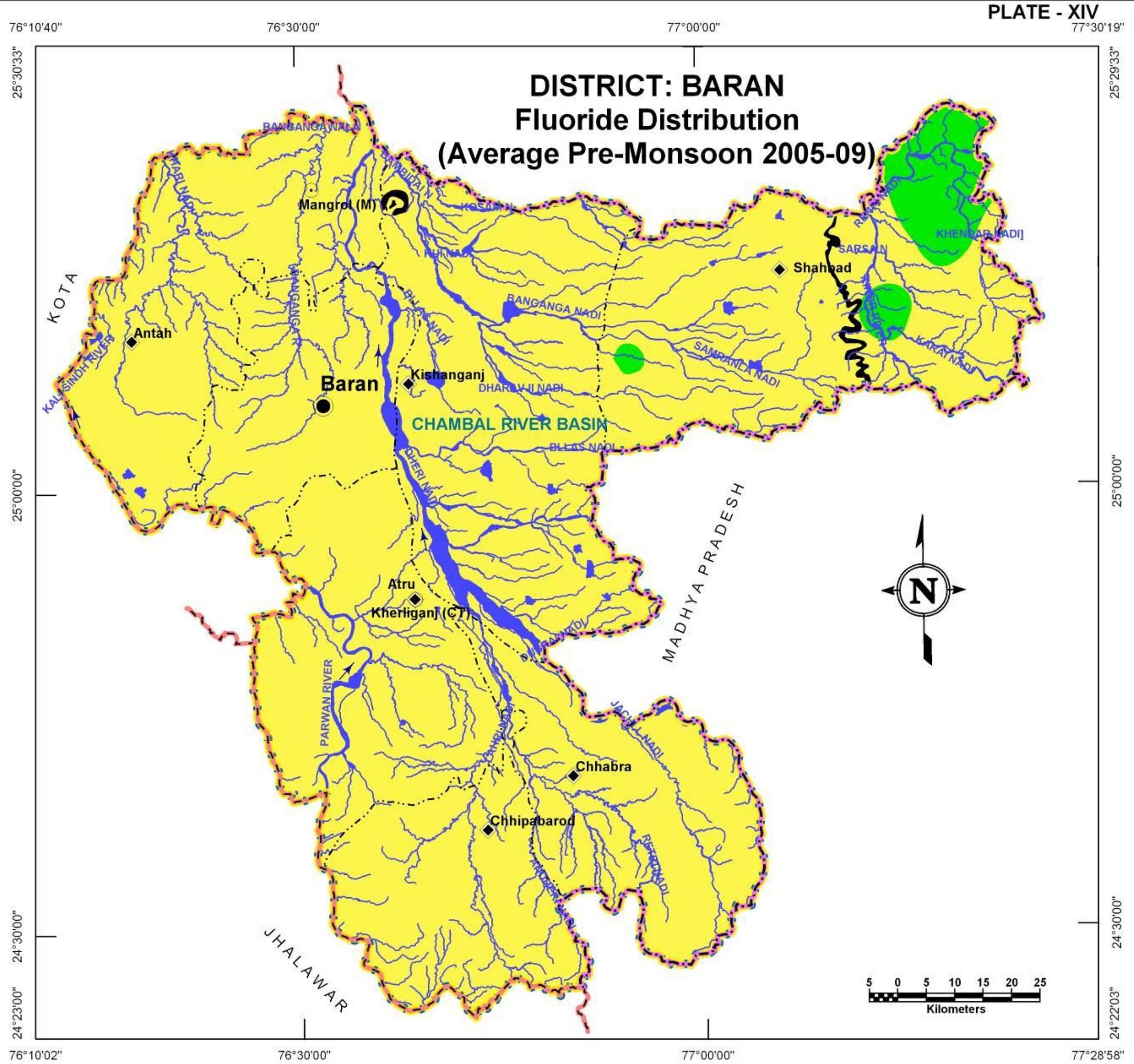
GROUND WATER NITRATE DISTRIBUTION

Plate XV shows distribution of Nitrate in ground water. Low nitrate concentration (<50 mg/l) is shown in yellow color and occupies approximately 77% of the district area which is suitable for agriculture purpose. The areas with moderately high nitrate concentration (50-100 mg/l) are shown in green color and occupy approximately 19% of the district area, largely in eastern part of Chhabra and around Kishanganj, Mangrol and Atru. That leaves only about 4% of the district area that has high nitrate concentration (>100 mg/l) which is shown in red colored patches, largely northern part of Atru and eastern part of Kishanganj where the ground water is not suitable for agriculture.

Table: Block wise area of Nitrate distribution

Nitrate concentration Range (mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)														Total Area (sq km)
	Antah		Atru		Baran		Chhabra		Chhipabarod		Kishanganj		Shahbad		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
<50	825.6	79.9	674.4	71.3	500.3	78.4	656.7	82.2	551.0	80.5	883.3	61.1	1,283.0	90.3	**5,374.3**
50-100	205.3	19.9	224.4	23.7	101.2	15.8	142.2	17.8	132.8	19.4	423.3	29.3	129.0	9.1	**1,358.2**
>100	1.8	0.2	47.4	5.0	37.3	5.8	-	-	0.4	0.1	138.1	9.6	8.4	0.6	**233.4**
Total	**1,032.7**	**100.0**	**946.2**	**100.0**	**638.8**	**100.0**	**798.9**	**100.0**	**684.2**	**100.0**	**1,444.7**	**100.0**	**1,420.4**	**100.0**	**6,965.9**

PLATE - XIV



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

Basin Boundary

Water Bodies:

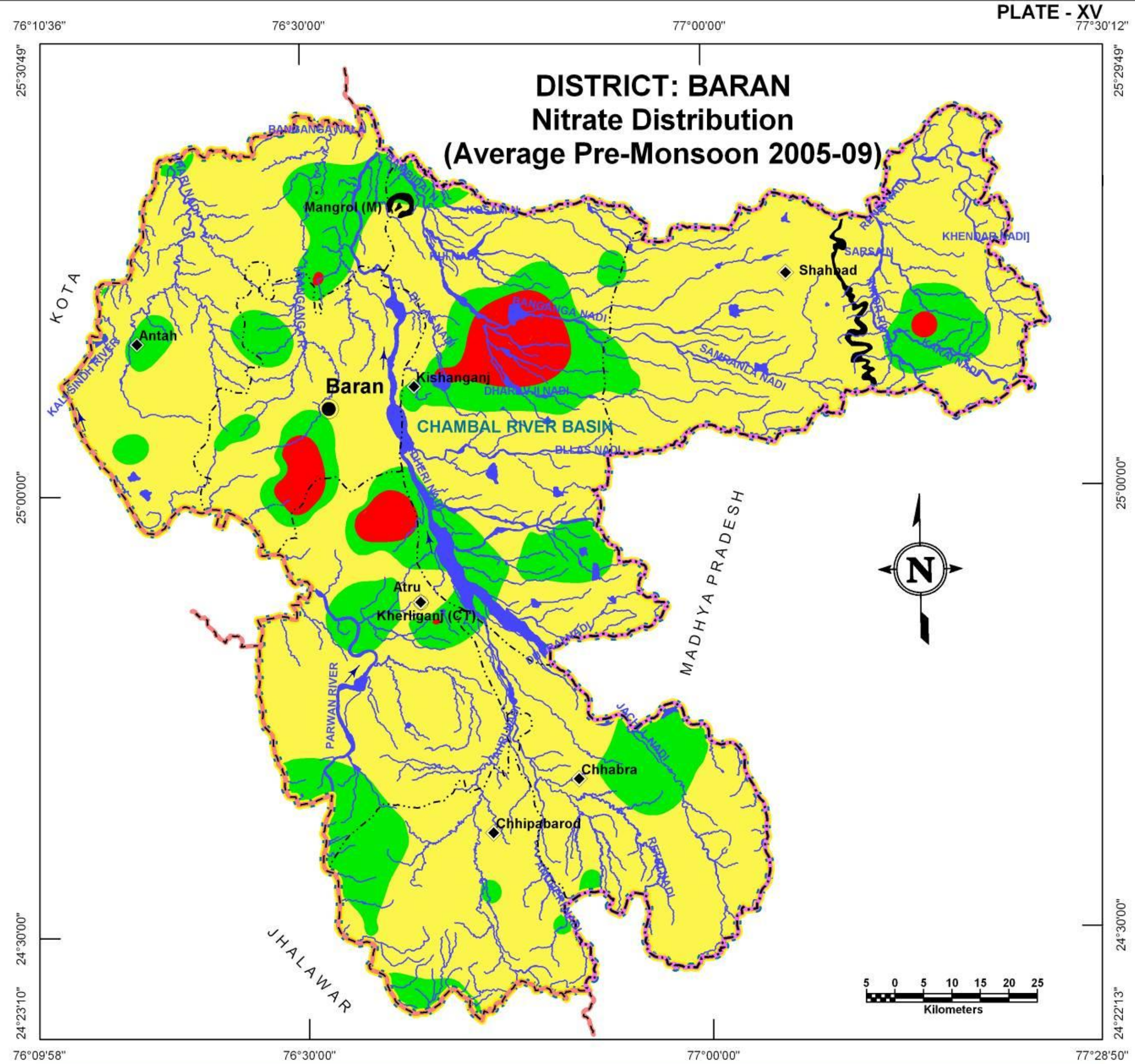
- River / Streams
- Ponds / Reservoirs

Hills

Fluoride Concentration (mg/l):

- < 1.5
- 1.5-3.0

PLATE - XV



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

Basin Boundary

Water Bodies:

- River / Streams
- Ponds / Reservoirs

Hills

Nitrate Concentration (mg/l):

- < 50
- 50-100
- >100

DEPTH TO BEDROCK

DISTRICT – BARAN

From hydrogeological perspective, the beginning of massive bedrock has been considered for defining top of bedrock surface. The major rocks types occurring in the district are Sandstone, Limestone, Shale and Basalt. Depth to bedrock map of Baran district (Plate – XVI) reveals that occurrence of bedrock at very shallow depth less than 20m below ground level in major part of the district excluding in the southwestern and northwestern parts of the district where the depth to bedrock varies from 20m bgl to 40m bgl. The maximum depth of bedrock (more than 40m bgl) found in Antru block.

Depth to bedrock (m bgl)	Block wise area coverage (sq km)														Total Area (sq km)
	Antah		Atru		Baran		Chhabra		Chhipabarod		Kishanganj		Shahbad		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
<20	519.0	50.3	205.2	21.7	603.3	94.4	612.4	76.7	192.2	28.0	1,239.6	85.8	1,420.4	100.0	**4,792.1**
20-40	513.7	49.7	712.5	75.3	35.5	5.6	186.5	23.3	492.0	72.0	205.1	14.2	-	-	**2,145.3**
>40	-	-	28.5	3.0	-	-	-	-	-	-	-	-	-	-	**28.5**
Total	**1,032.7**	**100.0**	**946.2**	**100.0**	**638.8**	**100.0**	**798.9**	**100.0**	**684.2**	**100.0**	**1,444.7**	**100.0**	**1,420.4**	**100.0**	**6,965.9**

UNCONFINED AQUIFER

Alluvial areas

Aquifer formed from alluvial materials covers only 327 sq km and mainly concentrated on Baran block and a very limited presence has observed in Antah and Atru block. The thickness of alluvial zone is less than 10 meter.

Unconfined aquifer Thickness (m)	District Area (sq km) coverage							Total Area (sq km)
	Antah	Atru	Baran	Chhabra	Chhipabarod	Kishanganj	Shahbad	
< 10	27.6	9.9	283.4	-	-	6.0	-	**326.9**
Total	**27.6**	**9.9**	**283.4**	**-**	**-**	**6.0**	**-**	**326.9**

Hardrock areas

Weathered, fractured and jointed rock formations occurring at shallower depths constitute good unconfined aquifers. Such zone ranges in thickness from less than 10 meter to around 20m covering almost entire parts of the district. Central part of Atru and a small sport in Antah block have weathered/fractured hardrock thickness upto 20m.

Unconfined aquifer Thickness (m)	District Area (sq km) coverage							Total Area (sq km)
	Antah	Atru	Baran	Chhabra	Chhipabarod	Kishanganj	Shahbad	
< 10	1,004.6	796.7	355.4	798.9	684.2	1,438.7	1,420.4	**6,498.9**
> 10	0.5	139.6	-	-	-	-	-	**140.1**
Total	**1,005.1**	**936.3**	**355.4**	**798.9**	**684.2**	**1,438.7**	**1,420.4**	**6,639.0**

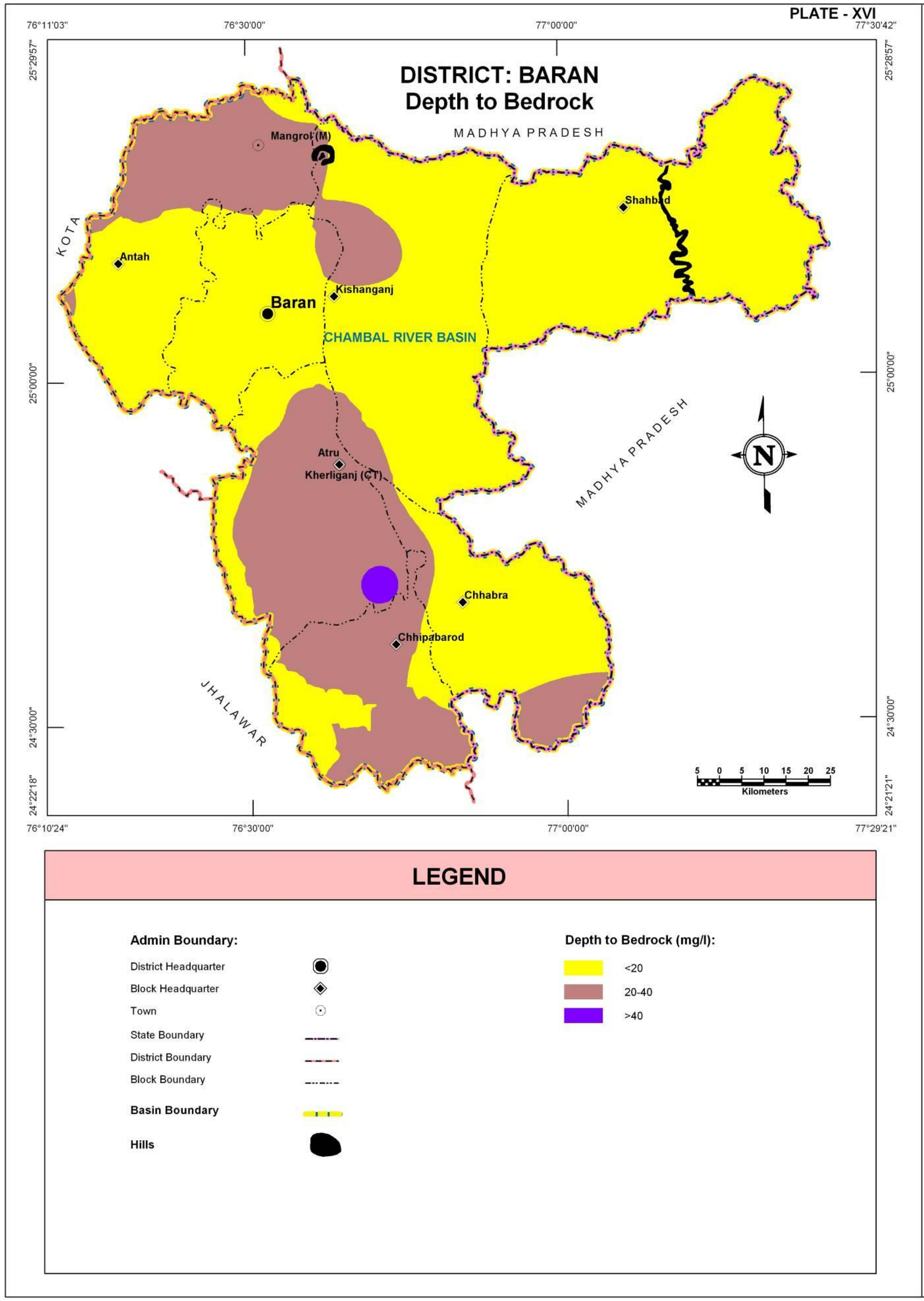


PLATE - XVI
DISTRICT: BARAN
Depth to Bedrock
MADHYA PRADESH
KOTA
JHALAWAR
Mangrol (M)
Shahbad
Antah
Baran
Kishanganj
CHAMBAL RIVER BASIN
Atru
Kherliganj (CT)
Chhabra
Chhipabarod
Kilometers
LEGEND
Admin Boundary:
District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary
Basin Boundary
Hills
Depth to Bedrock (mg/l):
<20
20-40
>40

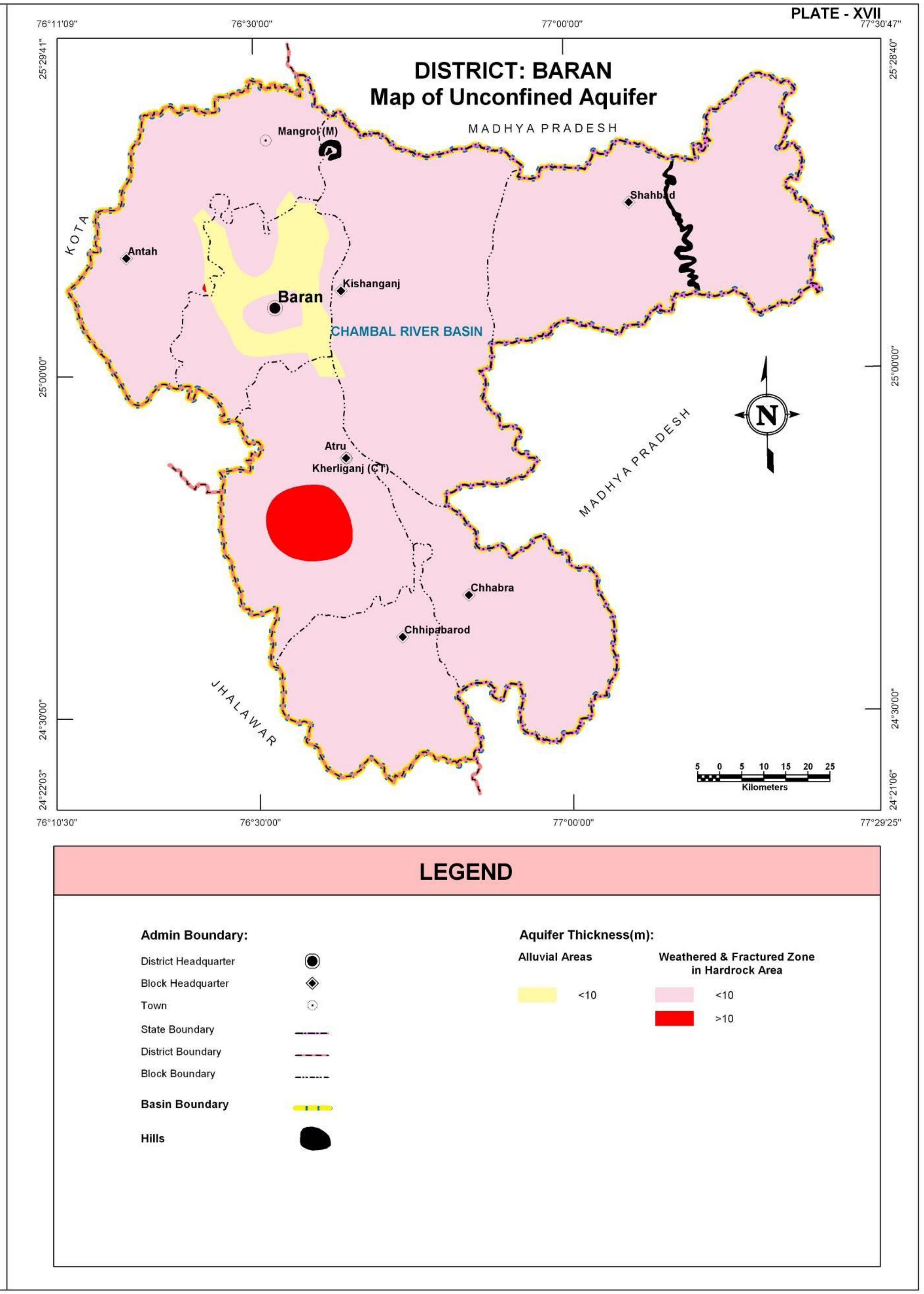


PLATE - XVII
DISTRICT: BARAN
Map of Unconfined Aquifer
MADHYA PRADESH
KOTA
JHALAWAR
Mangrol (M)
Shahbad
Antah
Baran
Kishanganj
CHAMBAL RIVER BASIN
Atru
Kherliganj (CT)
Chhabra
Chhipabarod
Kilometers
LEGEND
Admin Boundary:
District Headquarter
Block Headquarter
Town
State Boundary
District Boundary
Block Boundary
Basin Boundary
Hills
Aquifer Thickness(m):
Alluvial Areas
<10
Weathered & Fractured Zone
in Hardrock Area
<10
>10

Glossary of terms

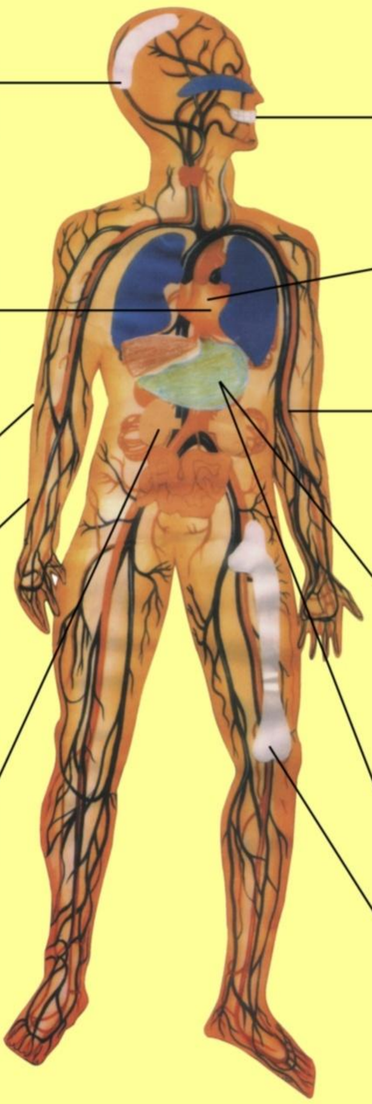
S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a ground water reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUND WATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity ($pH < 7$) or alkalinity ($pH > 7$).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from ground water without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after its complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits

(Contd…)

Ground Water Department,
Rajasthan

European Union
State Partnership Programme

भूजल में घुले मुख्य तत्वों की अधिकता का मानव शरीर पर दुष्प्रभाव



बोरोन-स्नायु तन्त्र पर प्रभाव

क्लोराइड-सोडियम के साथ मिलकर उच्च रक्त चाप

कैल्शियम-जोड़ों में कड़ापन

आर्सेनिक-त्वचा रोग, कैंसर

लेड-बच्चों के शारीरिक व मानसिक विकास में बाधा वयस्कों में गुर्दे के रोग

फ्लोराइड - दंत क्षरण

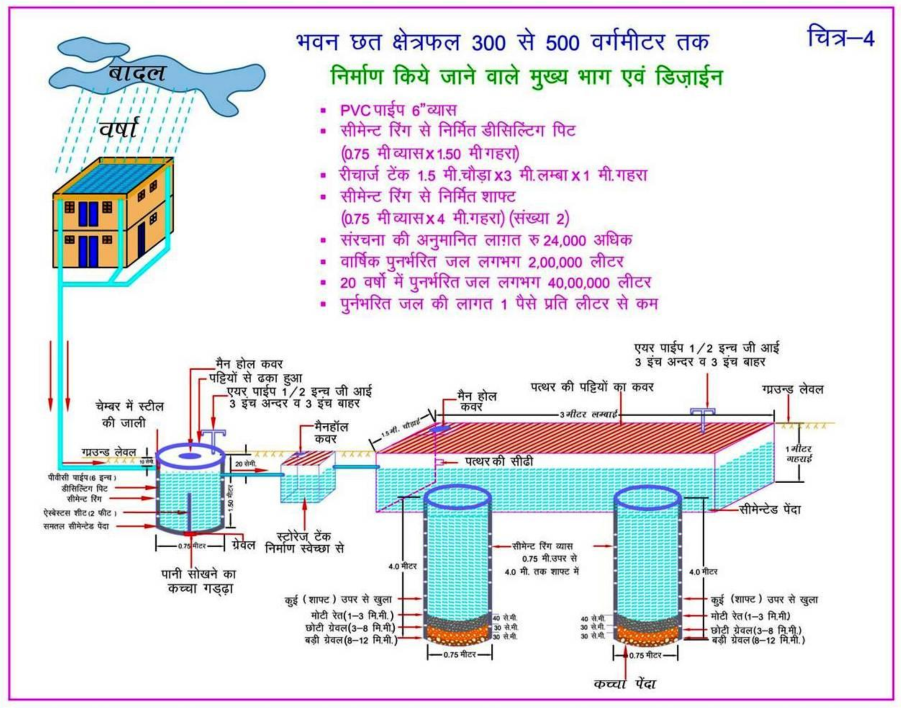
सोडियम-हृदय, गुर्दा व रक्त परिसंचरण रोगों से ग्रसित लोगों को हानिकारक

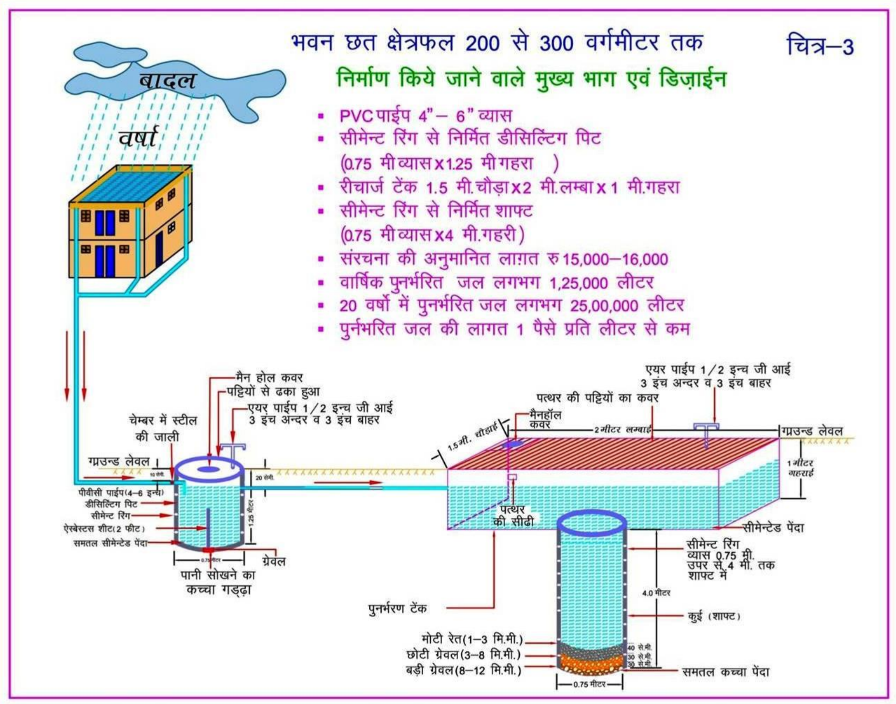
नाइट्रेट-नवजात शिशुओं में ब्लू बेबी बीमारी (मेथमोग्लोबिनिमिया)

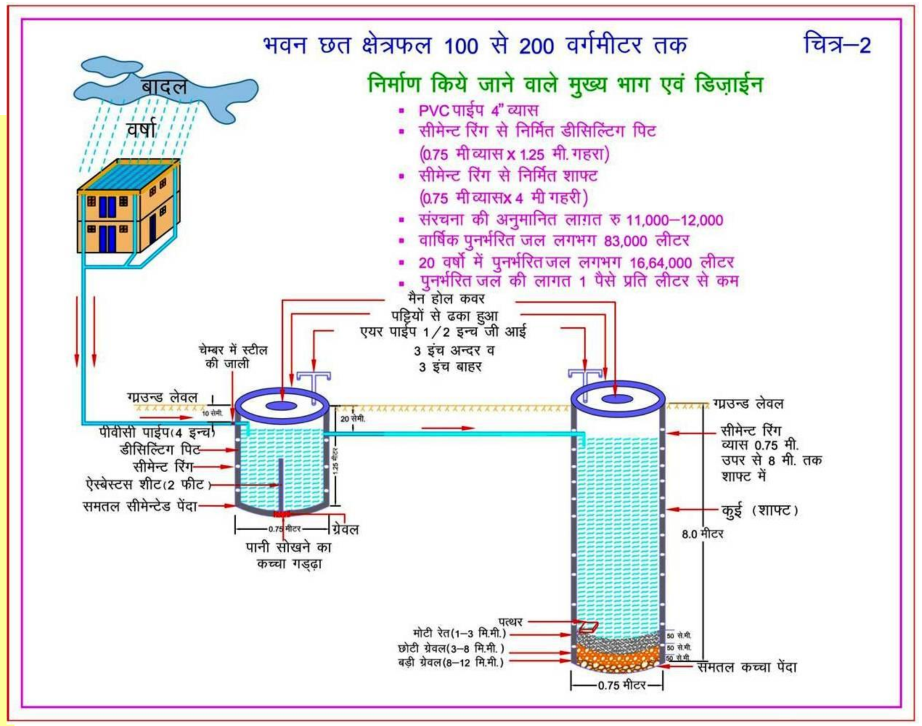
सल्फेट-अधिकता में मैग्नेशियम के साथ मिलकर दस्तावर

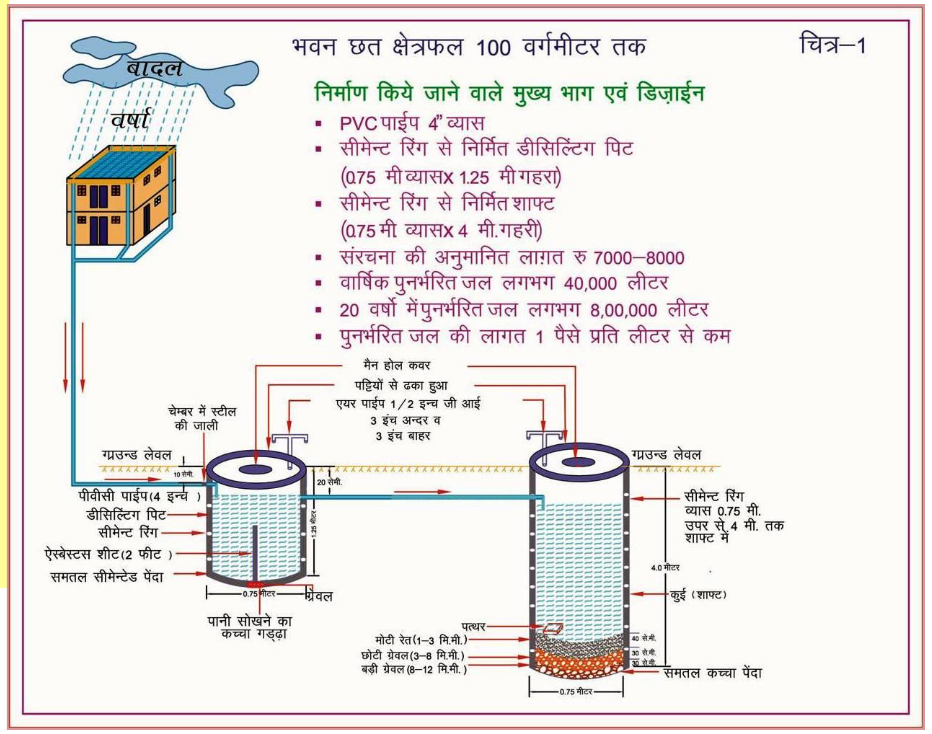
आयरन-आयरन जीवाणु से आमाशय संबंधी रोग

फ्लोराइड-जोड़ों में अकड़न, हड्डियों में मुड़ाव









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भूजल अमूल्य है इसे प्रदूषित न करें।

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Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoing wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water

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